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Wearable ECG Monitoring System with Explainable AI for Early Detection of Cardiac Anomalies in High-Risk Patients

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1 Introduction and Problem Validation

1.1 Background and Motivation

Cardiovascular diseases (CVDs) represent a primary health concern in Sri Lanka, carrying exceptionally high risks for patients with pre-existing conditions such as diabetes and hypertension. Despite the critical need for vigilant cardiac tracking, a significant disparity exists in healthcare infrastructure. Many rural hospitals and peripheral clinics completely lack essential electrocardiogram (ECG) devices due to the prohibitive cost of standard medical-grade equipment. Consequently, early screening and continuous diagnostic capabilities remain inaccessible to a large portion of the high-risk population.

1.2 Problem Statement

Through consultation with medical professionals¹ and field analysis, four critical gaps were identified in current cardiac diagnostic workflows:

- **Continuous Monitoring Gap:** Standard clinical ECG systems only capture short-duration snapshots (typically 5–10 seconds). These brief recordings are inherently incapable of capturing intermittent arrhythmias or tracking gradual waveform shifts over extended periods.
- **Diagnostic Misinterpretation Risks:** Acute cardiac events, such as myocardial infarctions, frequently present with atypical symptoms (e.g., epigastric pain). Without continuous tracking, these signs are easily misdiagnosed as benign gastrointestinal issues like gastritis, delaying life-saving interventions.
- **High Cognitive Load on Clinicians:** In understaffed and high-pressure emergency units, junior doctors bear the burden of manually analysing voluminous, paper-based ECG strips. This high cognitive load increases interpretation time and elevates the risk of human error.
- **Lack of Simultaneous Intelligence:** Existing affordable or portable diagnostic tools lack the dual capability to concurrently monitor cardiac signals and execute real-time anomaly detection.

1.3 Proposed Solution: IntelliWave

To address these infrastructure and diagnostic gaps, this project introduces **IntelliWave**—an affordable, ultra-portable and battery-powered 3-lead wearable ECG monitoring system. Optimized for day-to-day use by high-risk patients, the device bridges the continuous monitoring gap by logging long-term cardiac data.

Simultaneously, it integrates an embedded, explainable AI module powered by a Multi-Level Classification Framework. This software layer flags localized waveform anomalies (such as ST-segment deviations and irregular RR intervals) in real time. Rather than replacing clinical authority, the system operates strictly as an intelligent decision-support tool, substantially lowering the cognitive load on frontline medical staff and accelerating early, life-saving interventions.

¹The gaps identified in this workflow were formulated through professional consultation with Dr. Raleesha Niwanthaka, Medical Technology Unit (MTU), Faculty of Medicine, University of Moratuwa. The authors gratefully acknowledge his clinical insights.

2 General Product Definition and Market Analysis

2.1 Marketing Justification and Market Need

The concept for *IntelliWave* stems from a clear socio-economic and clinical gap in the South Asian healthcare market. While consumer smartwatches offer basic heart-rate tracking, they lack clinical validity, multi-lead morphology mapping and affordability for the masses. Conversely, hospital-grade Holter monitors are bulky, intrusive and structurally scarce in local healthcare systems.

From a market positioning standpoint, there is an urgent demand for an intermediate tier of medical devices: an affordable, accessible, prescriptive wearable ECG that bridges the gap between daily patient lifestyle and clinical-grade emergency triage. By targeting high-risk diabetic and hypertensive demographics, the marketing outlook identifies a sustainable, high-volume demand within both private and public healthcare sectors in Sri Lanka.

2.2 Benchmarking and Existing Product Analysis

To effectively position our design, the current state-of-the-art market solutions were analyzed:

- **iRhythm Zio Patch (International):** A single-lead continuous monitoring patch. While highly effective and low-profile, its single-use nature, dependence on premium cloud infrastructures and high cost make it entirely unviable for deployment in developing regions like Sri Lanka.
- **Traditional 3/5-Lead Holter Monitors (Local Clinical Standard):** Devices like the *Schiller Holter Systems* are widely used in local private hospitals. However, they require complex patient tethering, are uncomfortable for multi-day wear, do not offer real-time edge-AI anomaly detection and create massive data backlogs for overworked clinicians.

2.3 Literature and Media Context

Recent publications in medical electronics and local health reports emphasize a critical bottleneck: the ratio of cardiologists to cardiac patients in Sri Lanka is severely strained. According to regional healthcare summaries, over 30% of premature deaths are attributed to cardiovascular complications, often compounded by late detection. Media features in medical technology reviews emphasize that *Explainable AI (XAI)* in telehealth is the next mandatory frontier to reduce human error and alleviate the clinical cognitive load in crowded emergency triage centers.

2.4 User Profile and Market Segmentation

The target ecosystem for *IntelliWave* is split into two distinct segments:

1. **Primary End-User (The Patient):** Adults over 40 years of age diagnosed with hypertension, Type-2 diabetes, or transient arrhythmias. These users require a non-intrusive, single-button or automated device usable in day-to-day rural or urban routines without technical expertise.
2. **Secondary User (The Clinician):** Emergency unit junior doctors and medical officers who require immediate, interpreted, visual segment flags rather than raw, unorganized hours of ECG wave data.
3. **Market Segment:** Low-cost Remote Patient Monitoring (RPM) and Point-of-Care (PoC) diagnostic tools.

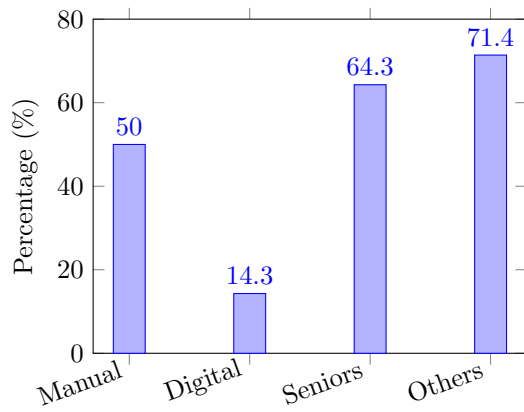


Figure 1: Current ECG interpretation methods.

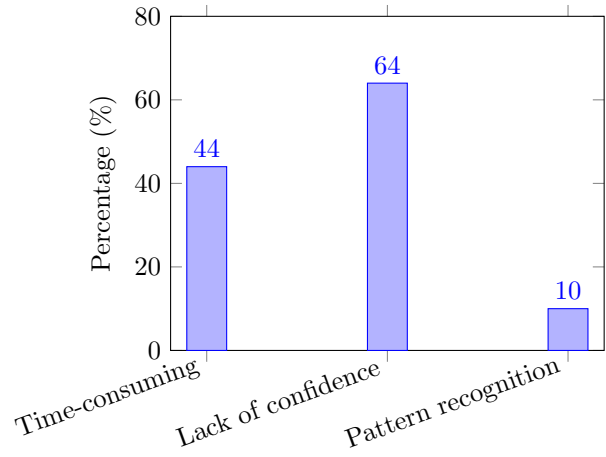


Figure 2: Challenges in ECG interpretation.

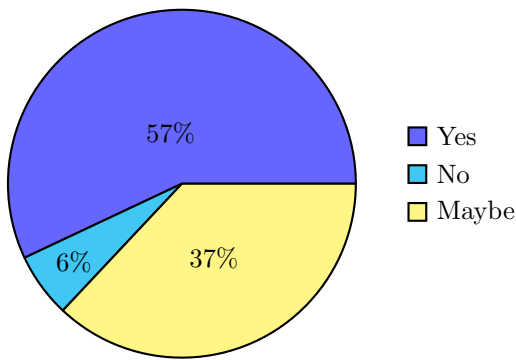


Figure 3: Interest in an AI-based ECG diagnostic tool.

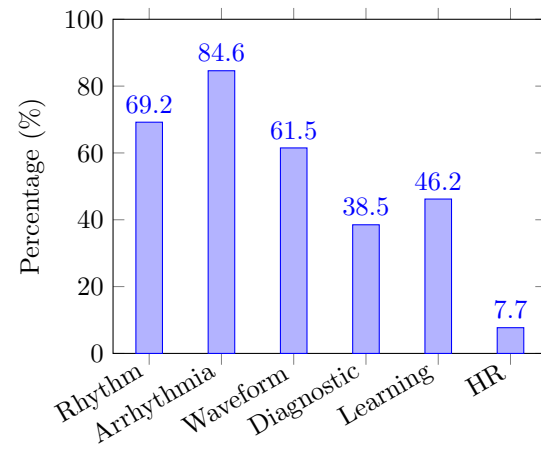


Figure 4: Desired features in an AI-assisted ECG tool.

2.5 Design Market Survey Insights

A preliminary market validation survey was conducted across medical faculties and clinical training centres in Sri Lanka, supplemented by consultations with medical experts. The core takeaways driving our design parameters include:

- **Usability:** 85% of respondents noted that high-risk patients drop out of long-term monitoring due to the discomfort of traditional skin-electrode wiring.
- **Diagnostic Bottleneck:** Junior clinicians confirmed that manual analysis of multi-hour ECG strips in busy wards takes an average of 10–15 minutes per patient, heavily validating the need for an automated AI anomaly flagging mechanism.

3 Engineering Design Architecture and Structural Mapping

Following the final lock of technical specifications, the development cycle shifts into foundational architectural design. This phase establishes the internal processing pathways, spatial constraints and external regulatory frameworks governing the *IntelliWave* system.

3.1 System Structure Diagram

The structural blueprint of the platform is partitioned into isolated Analog Front-End (AFE), Digital Processing, Power Distribution and Edge Interface domains. Figure 5 maps the programmatic and electrical signal path across the entire hardware-software ecosystem.

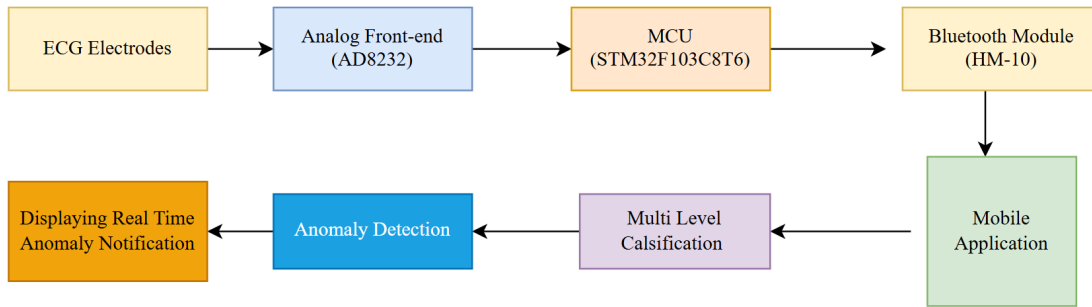


Figure 5: Block Diagram

3.2 Regulatory Standards Compliance Framework

To bridge the gap between laboratory bench prototypes and active hospital decision-support systems, *IntelliWave* incorporates hardware and architectural boundaries derived directly from global biomedical standards:

Table 1: Applicable Regulatory Frameworks and Design Mandates

Standard Reference	Scope and Regulatory Focus	Direct Implementation in Engineering Design
IEC 60601-1	General medical device electrical safety and performance requirements.	Complete optical or galvanic isolation between the active digital processor processing loop and analog bio-amplifier contact paths.
IEC 60601-2-47	Specific safety/performance bounds for ambulatory electrocardiographic systems.	Front-end configuration optimized to continuously preserve an input impedance $\geq 100\text{ M}\Omega$ and resolve baseline noise over a minimum 24-hour cycle.
ANSI/AAMI EC38	Standards for single-patient use and specialized ambulatory ECG systems.	Software integration of real-time sampling frequencies ($\geq 250\text{ SPS}$) mapping signal components without aliasing distortion.
IEC 60601-1-2	Electromagnetic compatibility (EMC) emissions and immunity limitations.	Deployment of local copper ground isolation traces, separate digital/analog PCB planes and input ferrite filtering blocks.

4 Electronics Design

4.1 Power Supply Circuit Design

The primary objective of the power supply sub-assembly is to regulate the variable battery voltage—ranging from 3.5 V to 4.2 V—down to a stable 3.3 V rail, ensuring minimum noise interference for sensitive analog front-end components.

Three design topologies were evaluated for this application:

1. **Direct Buck Regulation:** Direct step-down to 3.3 V using a buck converter.
2. **Direct Linear Regulation:** Direct regulation using a Low-Dropout (LDO) linear regulator.
3. **Hybrid Regulation:** A combined Buck-Boost converter followed by an LDO regulator.

4.1.1 Topology Evaluation and Trade-offs

Option 1: Direct Buck Converter While highly efficient, buck converters introduce significant high-frequency switching ripples into the power rail due to their periodic switching nature. Because the target system processes Electrocardiogram (ECG) signals, which are exceptionally weak (typically ranging from 0.5 mV to 5 mV), this high-frequency noise would severely degrade the Signal-to-Noise Ratio (SNR) of the analog front-end. Filtering this noise to an acceptable level adds unnecessary circuit complexity and footprint.

Option 2: Direct Low-Dropout (LDO) Regulator Typical LDOs exhibit a dropout voltage (ΔV) between 0.3 V and 1.5 V. For guaranteed 3.3 V regulation, the input battery voltage must remain strictly above $V_{\text{out}} + \Delta V$ (approximately 3.6 V to 3.7 V). Since a standard Lithium-ion battery discharges down to 3.5 V, a direct LDO topology would fail to maintain regulation during the latter half of the battery’s discharge cycle, severely limiting the operational lifespan of the device.

4.1.2 Selected Topology: Hybrid Buck-Boost with LDO

To balance both noise performance and battery life optimization, a hybrid topology consisting of a Buck-Boost converter cascaded with an LDO regulator was selected.

- **Stage 1 (Buck-Boost Converter):** This stage steps the fluctuating battery voltage (3.5 V to 4.2 V) up or down to a constant, stable intermediate rail of 3.7 V. This ensures full utilization of the battery’s capacity down to its safe cutoff voltage.
- **Stage 2 (LDO Regulator):** The LDO takes the stable 3.7 V intermediate input and steps it down to the final 3.3 V rail. Because the LDO possesses a high Power Supply Rejection Ratio (PSRR), it effectively suppresses the switching ripples generated by the preceding Buck-Boost stage, delivering an ultra-low-noise power rail ideal for ECG signal acquisition.

4.1.3 Component Selection, Detailed Design and Simulation Results

To implement the selected hybrid topology, specific integrated circuits (ICs) were chosen and verified via parametric simulation models under full load conditions.

Buck-Boost Stage: TPS63900DSKR The first stage utilizes the **TPS63900DSKR**, a high-efficiency synchronous buck-boost converter. It is configured to regulate the varying battery voltage down to a stable intermediate rail of 3.7 V. Simulation metrics under peak operating conditions ($I_{\text{out}} = 400$ mA, worst-case minimum input $V_{\text{in}} = 3.0$ V) yield the following performance characteristics:

- **Efficiency and Power Delivery:** The converter operates at a switching frequency of 1.1 MHz, delivering an output power (P_{out}) of 1.48 W with a steady-state efficiency of 90.6 %.

- **Thermal and Power Dissipation:** The total power dissipation (P_d) of this stage is 153.554 mW, where the IC accounts for 138.73 mW and the power inductor (L) accounts for 14.825 mW. Under these conditions, the IC junction temperature (T_j) stabilizes safely at 53.962 °C.
- **Regulation Quality:** The achieved operational output voltage is 3.7 V with a tight strict component-level voltage tolerance of 135.135 m%.
- **Physical Footprint:** The total Bill of Materials (BOM) count for this sub-assembly is 6 components, occupying a highly compact total footprint area of 47.0 mm².

LDO Regulation Stage: TPS7A4700RGWR For the final ultra-low-noise regulation layer, the **TPS7A4700RGWR** positive voltage linear regulator is cascaded directly from the 3.7 V intermediate rail. Simulation benchmarks at the target operating point ($V_{in} = 3.75$ V, $V_{out} = 3.3$ V, $I_{out} = 300$ mA) demonstrate superior noise rejection and thermal stability:

- **Noise and Ripple Rejection:** With a 100 kHz input source ripple of 40.0 mV_{p-p} representing residual switching noise, the LDO exhibits an estimated Power Supply Rejection Ratio (PSRR) of -60.0 dB. This actively dampens the peak-to-peak output ripple voltage down to an ultra-low 40.0 μ V_{p-p}, yielding a total integrated RMS output noise of just 4.273 μ V_{RMS}. This ultra-clean rail is critical for preserving the SNR of the weak (0.5 mV to 5 mV) analog ECG signals.
- **Thermal and Power Metrics:** The LDO runs at a steady-state efficiency of 87.223 %, outputting 990.0 mW of clean power. The total power dissipation within the linear stage is 145.023 mW. Given the IC's junction-to-ambient thermal resistance (θ_{JA}) of 32.5 °C W⁻¹, the junction temperature rises minimally, stabilizing at an optimal 49.713 °C. The IC ground current (I_{ground}) is clocked at 2.673 mA.
- **Physical Integration:** The LDO network achieves an exact target output of 3.3 V with a cumulative tolerance of 1.061 %. The stage consists of a 4-component BOM taking up a footprint of 60.0 mm².

4.1.4 Schematic Integration and Layout Considerations

To translate these high-performance simulation benchmarks onto physical hardware without sacrificing the simulated ultra-low-noise threshold, strict PCB layout guidelines must be enforced:

1. **Stage Isolation:** The high-frequency switching loop of the TPS63900 (operating at 1.1 MHz) along with its 14.825 mW power inductor path must be strictly isolated geographically from the input pins of the TPS7A4700 to mitigate radiated electromagnetic interference (EMI).
2. **Grounding Strategy:** A split ground plane architecture or single-point star grounding must be implemented. This isolates the high $I_{in, Avg}$ (544.52 mA) return path of the switching regulator from the clean analog ground plane dedicated to the LDO reference and the sensitive analog front-end components.

4.2 Core Processing Unit (MCU): STM32F103C8T6

The selection of the processing element required balancing robust processing throughput, strict deterministic timing for high-fidelity analog ADC sampling, wide development toolchain support and high cost-effectiveness for wearable biomedical device prototyping.

Strong Recommendation Rationale: While the ESP32 delivers integrated RF capabilities, its dual-core architecture introduces complex RTOS scheduling jitter and its high active current consumption induces significant thermal dissipation variations. This heat can threaten the baseline thermal stability of highly sensitive adjacent biopotential analog grounds like the AD8232 front-end. The Nordic nRF52832 offers competitive low-power operations, but severe supply chain logistics

Table 2: Parametric Trade-off Matrix for System Microcontrollers

Metric	ESP32-WROOM-32E	nRF52832-QFAA	STM32F103C8T6 (Selected)
Core	Dual-core Xtensa LX6	Single-core ARM Cortex-M4	Single-core ARM Cortex-M3
Ecosystem	Extensive (IoT)	Moderate (Wireless)	Exceptional (IDE & Community)
Availability	Highly Available	Severe Bottleneck (>8 wk lead)	Globally Abundant (Immediate)
Power	High (100–240 mA)	Ultra-low (<15 mA)	Low (15–50 mA)

and restrictive toolchain licensing present substantial barriers within rigid academic or commercial timeline constraints.

The **STM32F103C8T6** was selected as the optimal processing platform for this system. Built on a 32-bit ARM Cortex-M3 core operating at up to 72 MHz, it provides a highly deterministic Nested Vectored Interrupt Controller (NVIC), which is critical for executing precise, fixed-interval analog-to-digital conversions (ADC) required for accurate ECG waveform sampling. It features a high-performance internal 12-bit SAR ADC peripheral capable of directly capturing the conditioned analog output of the AD8232 without processing lag. Additionally, its robust hardware timers, low-power sleep modes and rich peripheral arrays (SPI, I2C, USART) allow for seamless communication with auxiliary modules while eliminating the architectural complexity and firmware overhead associated with complex wireless-first SoCs.

4.3 Wireless Connectivity: HM-10 BLE 4.0 Module

To enable real-time, wireless transmission of the conditioned ECG telemetry to a host device while maintaining low system complexity and strict power limits, the HM-10 Bluetooth Low Energy (BLE) module was selected.

Module Integration and Rationale: The HM-10 is an optimized, cost-effective peripheral subsystem based on the Texas Instruments CC2541 Bluetooth SoC. It communicates with the host STM32F103C8T6 microcontroller via a standard universal asynchronous receiver-transmitter (UART) interface using simple AT commands. This decoupling offers significant architectural advantages for a mixed-signal biomedical device:

- **Firmware Decoupling:** By running the entire BLE protocol stack onboard its own silicon, the HM-10 removes all wireless state-machine and connection-management overhead from the host MCU. This ensures that the STM32’s processing cycles remain fully deterministic, eliminating execution jitter during high-priority internal ADC sampling intervals of the AD8232 signal.
- **Electrical and RF Isolation:** Operating at a standard transmission power of 0 dBm, the module draws an exceptionally lean active current profile (~8.5 mA during RF transceiving). This significantly limits transceiving power spikes compared to high-power Wi-Fi/Bluetooth combos, preventing dynamic voltage drops across the system power distribution network (PDN).
- **PCB Layout Compatibility:** The module features a standardized edge-castellated form factor with an onboard PCB inverted-F antenna (IFA). This allows it to be placed at the far edge of the board layout, simplifying the implementation of a dedicated copper keep-out zone beneath the antenna to ensure efficient RF radiation without polluting the sensitive analog grounds of the AD8232 front-end.

4.4 Analog Front-End (AFE): AD8232

Preserving the microvolt-level structural components of an ECG complex (*P*-wave, *QRS* complex, *T*-wave) requires dedicated instrumentation optimized for biopotential signal conditioning and robust common-mode noise rejection.

Table 3: Parametric Comparison of Candidate ECG Analog Front-Ends

Parameter	TI ADS1292 (Integrated Digital AFE)	ADI AD8232 (Selected Analog AFE)
Output Interface	Digital output via SPI, ensuring noise-immune serial communication	Analog voltage output, requiring strategic component placement to minimize PCB trace noise coupling
Signal Conversion	Integrated 24-bit delta-sigma ADC, 2 channels for synchronous multi-lead tracking	Relies on the host MCU’s internal ADC for flexible, direct digitization
Firmware Complexity	Higher — requires SPI driver development, complex register configuration and custom digital filtering layers	Lower — allows instant, low-overhead sampling via standard MCU analog peripherals
Power Consumption	Optimized micro-power footprint (335 μ W/channel) but features higher registers-management overhead	Minimal power consumption (170 μ A typical active current), highly optimized for wearable topologies
Right-Leg Drive	Integrated dedicated RLD amplifier loop with internal routing matrix	Integrated dedicated RLD amplifier loop for active common-mode rejection
Footprint Area	Compact integration (5 mm $\times$ 5 mm TQFP or 4 mm $\times$ 4 mm QFN)	Minimal space constraints (4 mm $\times$ 4 mm LFCSP), maximizing layout flexibility for ultra-compact boards

Strong Recommendation Rationale: While the ADS1292 offers high-resolution digital conversion on-chip, its integration demands substantial firmware overhead, including dedicated SPI driver architectures, multi-register initialization sequences and extensive software-side digital filtering. This complexity increases processing latency and power consumption during active communication cycles, which can constrain lightweight, real-time embedded anomaly detection loops running continuously on constrained wearable hardware.

The **AD8232** was selected as the optimal architecture for this system. It provides a highly specialized, single-chip solution for biopotential signal conditioning, seamlessly combining an instrumentation amplifier, an operational amplifier and a right-leg drive (RLD) loop. It features an integrated high-pass filter that effectively eliminates baseline wander caused by motion artifacts and electrode polarization right at the analog stage, preventing ADC saturation. By outputting a fully conditioned analog signal directly to the host MCU’s high-resolution internal ADC, the system achieves a highly responsive, low-latency sampling path with minimal firmware complexity and reduced active power consumption. Layout vulnerabilities are mitigated by keeping the AD8232 in close physical proximity to the MCU’s ADC input pins, shielded by dedicated ground planes to ensure high signal fidelity.

4.5 Power Management Integrated Circuit (PMIC): BQ24074RGTR

Medical wearables must manage power-path distribution safely to prevent battery degradation or user-facing thermal overstress.

Strong Recommendation Rationale: Basic chargers like the TP4056 connect the system load directly in parallel with the battery cell during a charge cycle. This prevents the charger from correctly resolving the termination current threshold, keeping the lithium cell in a continuous high-voltage state that accelerates chemical degradation and raises thermal risks.

The **BQ24074RGTR** features automated Dynamic Power-Path Management (DPPM). If the input power source lacks sufficient capacity to supply the complete system load, the device automat-

Table 4: Battery Charger Architecture Breakdown

Feature	TP4056 (Standard Charger)	TI BQ24074RGTR (Selected Dynamic PMIC)
Architecture	Basic Linear Charger	Dynamic Power-Path Management (DPPM) Linear
System Power	Shares node with battery charging	Independent system load from battery charging
Thermal Control	None (Susceptible to thermal runaway)	Integrated thermal regulation loop (105 °C limit)
Safety Diagnostics	Basic LED Status	Input Overvoltage (up to 28 V), JEITA Thermistor

ically modulates the charge current or invokes a supplement mode to draw current from the battery. This feature enables dependable operation and safe charging, even from noisy or current-limited USB sources.

4.6 Battery Fuel Gauge: MAX17048G+T10

Accurately alerting a patient to a low-battery state requires reliable state-of-charge calculation.

Table 5: Fuel Gauge Methodology Comparison

Parameter	TI BQ27441-G1 (Coulomb Counter)	Maxim MAX17048G+T10 (Selected Gauge)
Sensing Element	External Low-Value Sense Resistor	No sense resistor required (V_{cell} tracking)
PCB Parasitics	High insertion loss at sense resistor	Zero insertion loss
Calibration	Complex multi-step calibration phase	Plugs into I2C bus with zero configuration
Quiescent Current	90.0 μA active	23.0 μA active (4.0 μA sleep)

Strong Recommendation Rationale: Conventional Coulomb-counting architectures like the BQ27441-G1 introduce an inline sense resistor into the power return path. This component generates constant I^2R power dissipation and shifts the system ground reference dynamically with current transients.

Additionally, Coulomb counters suffer from offset drift accumulation over time, requiring periodic full-to-empty cycles to re-calibrate. The **MAX17048G+T10** uses a proprietary ModelGauge algorithm that maps the cell voltage directly against dynamic electrochemical impedance models. This approach provides an exceptionally accurate State of Charge (SoC) metric without requiring a sense resistor, keeping the design layout clean and minimizing current consumption down to a low 23.0 μA .

4.7 Schematic Integration and Layout Considerations

To translate these high-performance simulation benchmarks onto physical hardware without sacrificing the simulated ultra-low-noise threshold, strict PCB layout guidelines must be enforced:

1. **Stage Isolation:** The high-frequency switching loop of the TPS63900 (operating at 1.1 MHz) along with its 14.825 mW power inductor path must be strictly isolated geographically from the input pins of the TPS7A4700 to mitigate radiated electromagnetic interference (EMI).

2. **Grounding Strategy:** A split ground plane architecture or single-point star grounding must be implemented. This isolates the high $I_{in, Avg}$ (544.52 mA) return path of the switching regulator from the clean analog ground plane dedicated to the LDO reference and the sensitive analog front-end components.

4.8 PCB Layer Stackup, Grounding Strategy & Schematics

4.8.1 Selected 4-Layer Stackup

To physically realize the noise isolation and signal integrity required for high-fidelity biopotential conditioning and high-speed microcontroller operation, a 4-layer stackup was selected and configured within the Altium Designer Layer Stack Manager. The plane assignment is structurally configured as follows: Layer 1 is dedicated to primary signal and critical analog routing, Layer 2 is a solid ground (GND) plane serving as the primary return path, Layer 3 carries the split analog and digital supply rails (AVDD + DVDD) as a shared power plane and Layer 4 is a secondary ground (GND) plane.

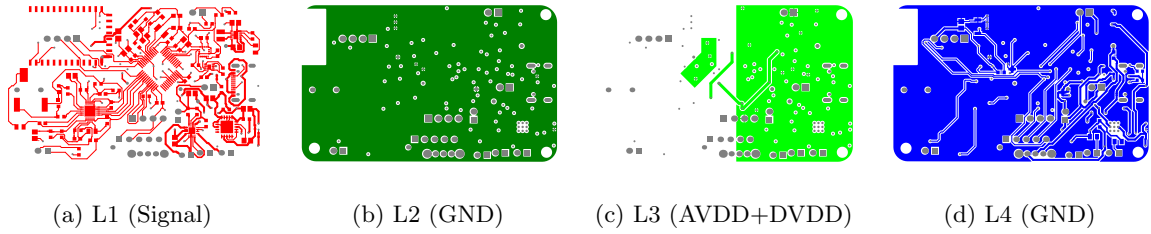


Figure 6: PCB Layers Comparison

Table 6: 4-Layer PCB Physical Stackup Summary

Layer	Type / Assignment	Material / Core	Thickness (mil)	Weight (oz)
1	Signal (Top)	Copper Foil (CF-004)	1.378	1.0
—	Dielectric 2 (Prepreg)	PP-006 (1080)	3.913	—
2	GND (Plane)	Copper Foil (CF-003)	0.598	0.5
—	Dielectric 1 (Core)	FR-4	49.803	—
3	AVDD / DVDD (Split Power Plane)	Copper Foil (CF-003)	0.598	0.5
—	Dielectric 3 (Prepreg)	PP-006 (1080)	3.913	—
4	GND (Bottom Plane)	Copper Foil (CF-004)	1.378	1.0
Total	Finished Thickness		61.981 mil	

The stack is dominated by a 49.803 mil FR-4 core sandwiched between the internal Layer 2 GND and Layer 3 AVDD+DVDD planes, giving a total build thickness of approximately 62 mil (~1.58 mm). This configuration forms an tightly coupled power-ground sandwich across the core to maximize inter-plane capacitance, helping to damp high-frequency power distribution network (PDN) noise. All four layers are interconnected through standard through-hole via structures spanning Layer 1 to Layer 4, ensuring reliable and cost-effective structural transitions for the STM32F103C8T6 escape routing and peripheral connections.

4.8.2 Rationale and Grounding Strategy

Placing a continuous, solid ground plane on Layer 2 immediately beneath the top signal layer minimizes the loop area for high-frequency digital current returns and sensitive analog pathways. This directly reduces radiated electromagnetic interference (EMI) and cross-channel noise pickup. Layer 2

acts as an unbroken, low-impedance reference plane directly under the AD8232 instrumentation amplifier front-end, protecting the low-amplitude, microvolt-level biopotential input lines from digital switching currents.

For Layer 3, mixed-signal routing optimization dictates isolating the analog and digital supply domains to prevent the high-frequency digital switching harmonics generated by the 72 MHz STM32F103C8T6 clock from corrupting the delicate analog conditioning path. The Layer 3 copper layer is strategically split into a localized AV_{DD} pour (feeding the AD8232) and a DV_{DD} pour (powering the MCU core and auxiliary peripherals).

Layer 4 acts as a secondary solid ground (GND) plane. This establishes a continuous return path for components mounted or routed on the bottom layer, shields the internal power planes from external environmental noise and clamps any residual EMI escaping the high-speed digital lines running across the board.

Strong Recommendation Rationale: To preserve electrical isolation and maintain signal integrity across this 4-layer topology during layout design:

- No signal trace should cross the AV_{DD}/DV_{DD} split gap on Layer 3 without a continuous reference plane directly adjacent on Layer 2 or Layer 4 to avoid generating large slot-antenna return loops that distort signals and radiate EMI.
- The sensitive analog input paths leading into the AD8232 (RA, LA, LL electrode traces) must be geographically isolated from high-speed digital buses (such as the MCU's USART, SPI, or SWD lines) to mitigate capacitive or inductive crosstalk into the microvolt-level ECG signals.
- Place 0.1 μF ceramic decoupling capacitors as physically close as possible to the supply pins of both the AD8232 and the STM32F103C8T6 to ground, ensuring low-impedance high-frequency AC return loops right at the silicon boundaries.
- Avoid routing long analog traces across the board; the analog output signal from the AD8232 should trace a short, direct path to the host MCU's internal 12-bit SAR ADC input pin to minimize environmental noise coupling.
- Explicitly map, name and constrain the Layer 3 power plane splits inside the Altium Designer Design Rules Check (DRC) system to ensure net-classes maintain proper electrical clearance before manufacturing.

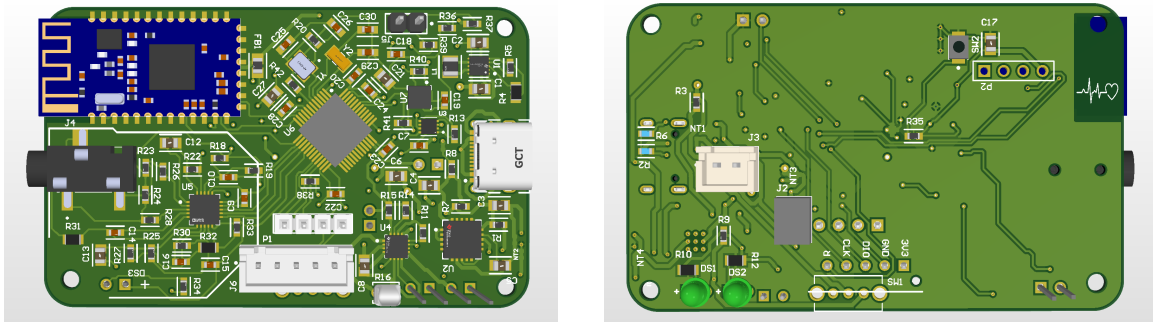
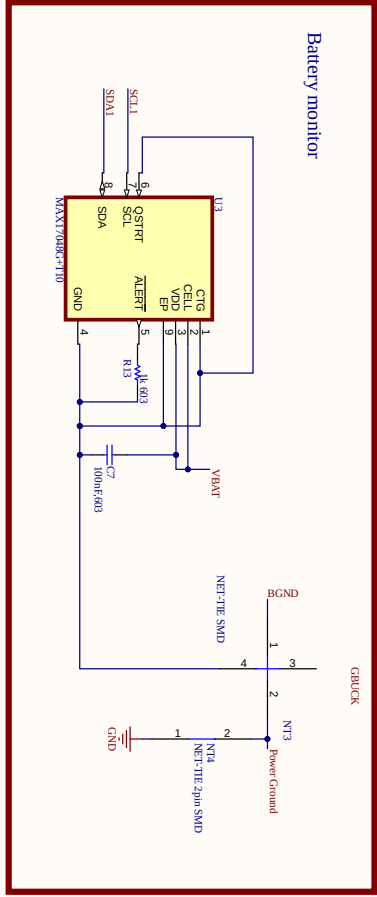
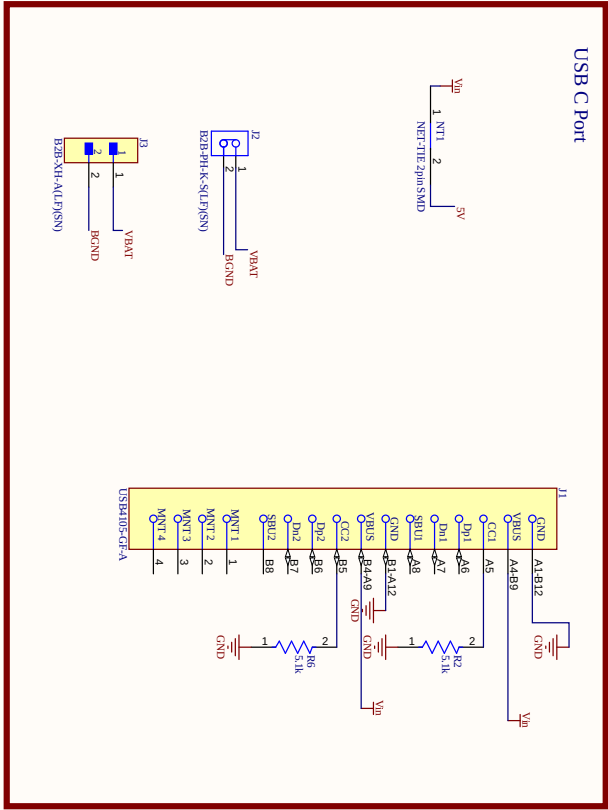
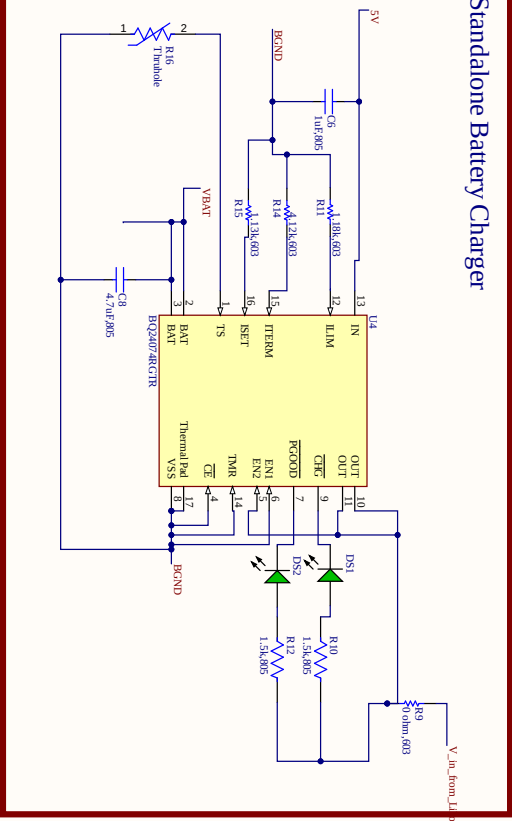
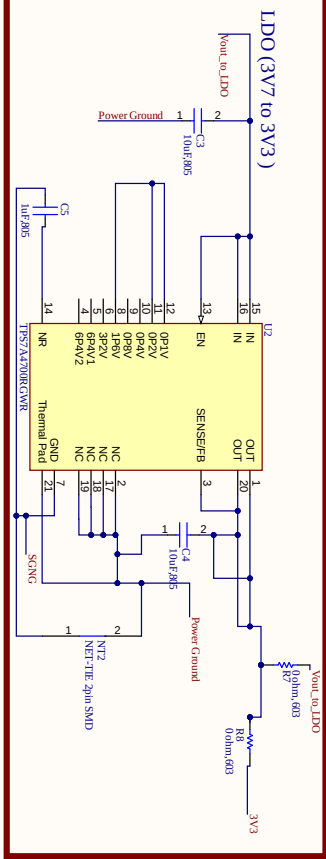
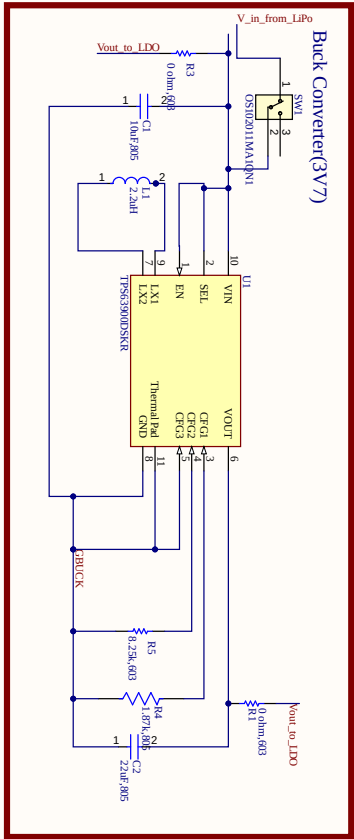


Figure 7: PCB Artwork: Top View (Left) and Bottom View (Right)

Power Management

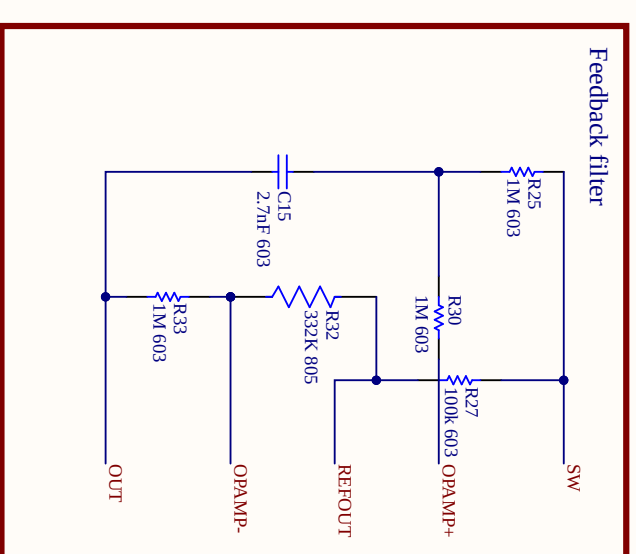
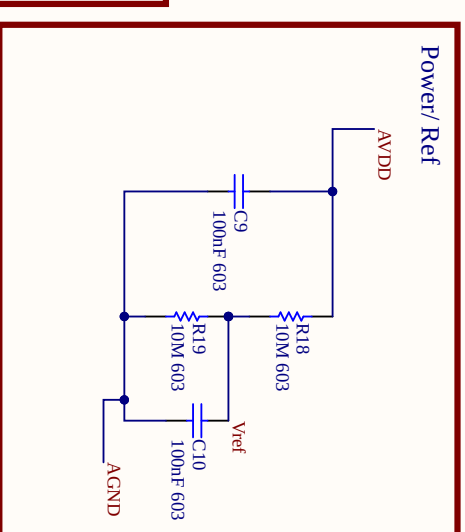
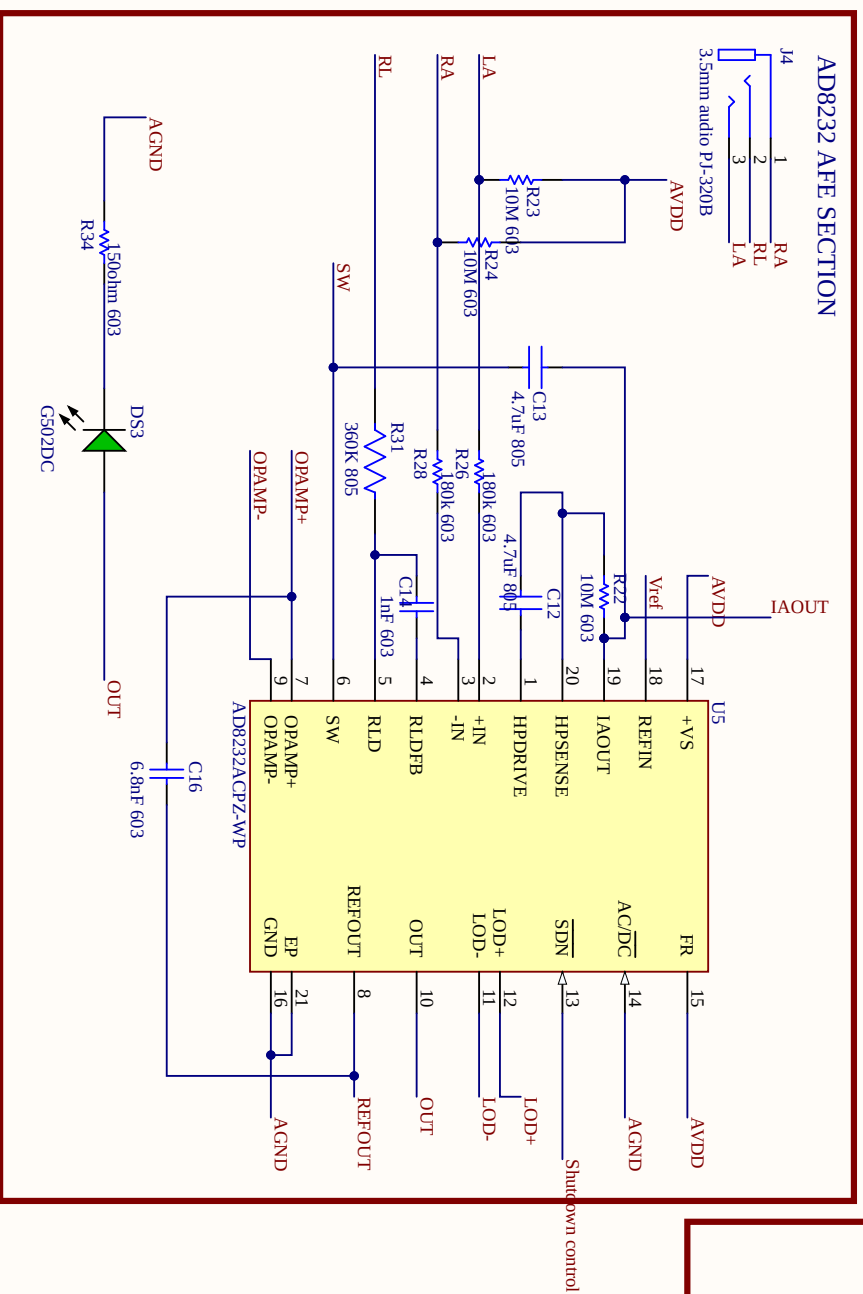


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FILE NAME	SCALE	ECG power mgmt
PCB DWG:	SCALE	ECG power mgmt

Altium

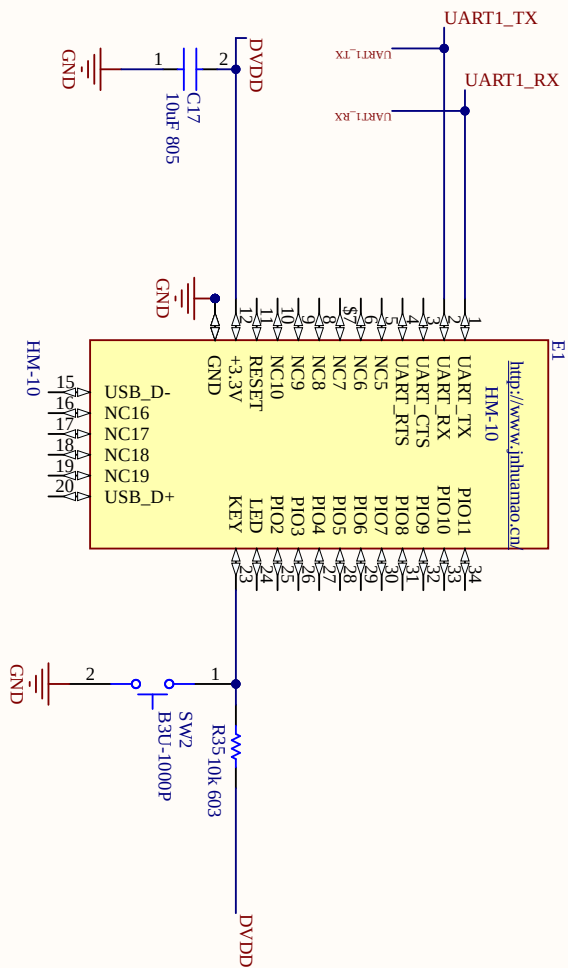
ECG power mgmt

Analog Front-End



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5 Industrial (Mechanical) Design

5.1 Preliminary Sketches

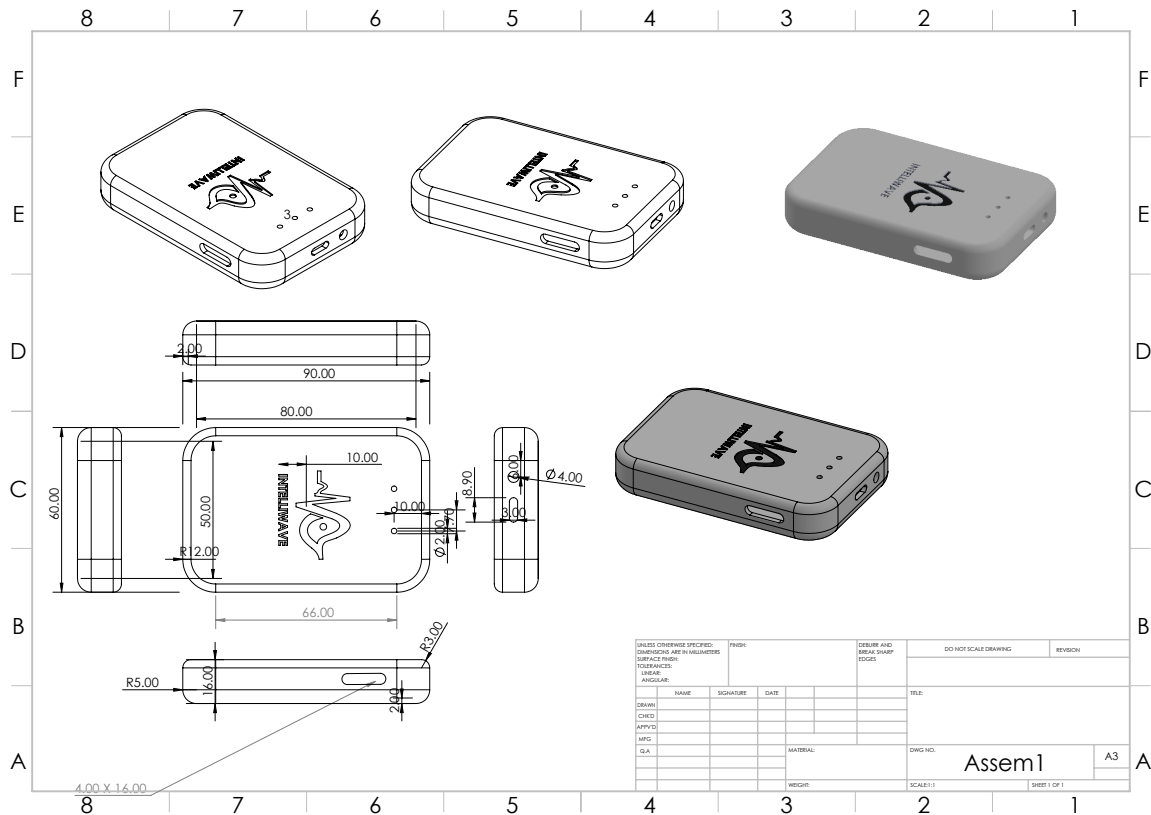


Figure 8: Preliminary Sketch 1

5.2 Design Iterations: Rationale for Selection, Non-Selection and Changes

To optimize ergonomics, durability and user experience, the system's mechanical enclosure underwent significant structural refinements between the first and second design iterations. The specific modifications made to the physical chassis are detailed below:

1. **Integrated Strap Holder Mechanism:** In the first iteration, mounting the device to the wearable strap was cumbersome. The second iteration introduces a dedicated band holder system, allowing users to effortlessly connect and detach the main monitoring housing from the chest band.
2. **Ergonomic Switch Protection Cover:** The initial exposed cutout for the power toggle compromised the device's aesthetics and left internal components vulnerable to environmental ingress. A flush switch cover was designed for the second iteration to completely mask the structural chassis gap while providing improved tactile feedback and user experience.
3. **Chassis Footprint Optimization:** The first-generation enclosure was bulky and unsuited for long-term wearable use. By packing internal components tighter and optimizing component placement, the overall enclosure dimensions were reduced in the second iteration to achieve a highly compact, handheld and ergonomic form factor.
4. **Port Alignment and Mounting Tolerances:** The mounting apertures for the 3.5 mm biometric sensor jack and the USB Type-C charging interface were recessed and structurally

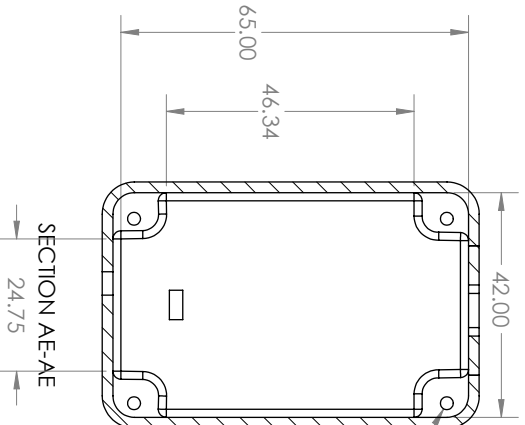
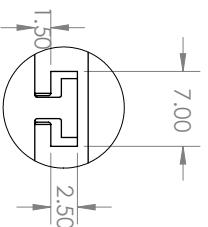
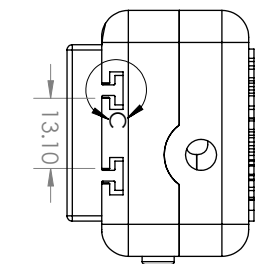
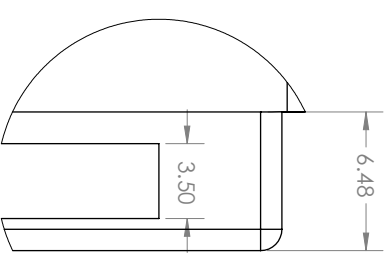
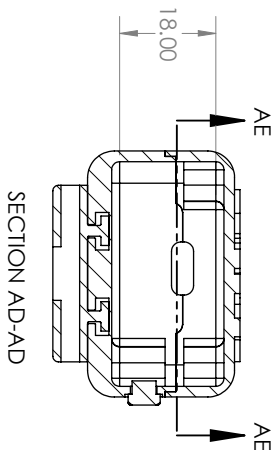
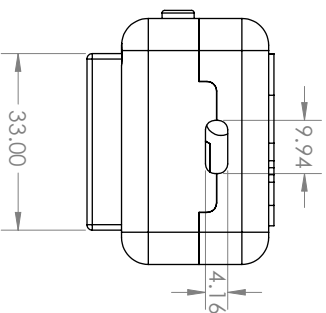
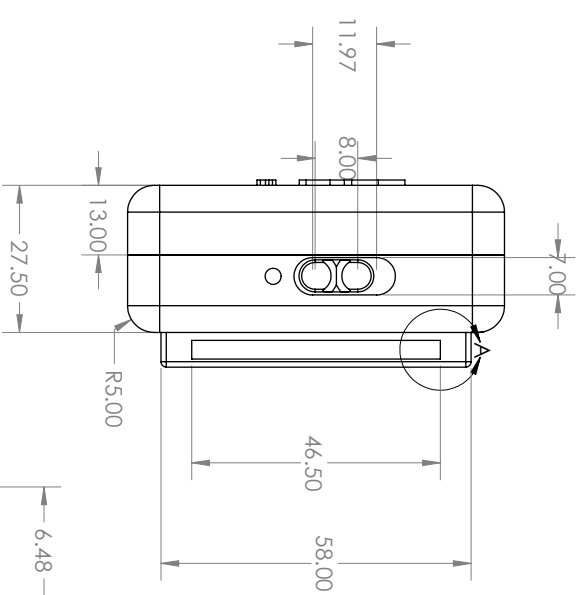
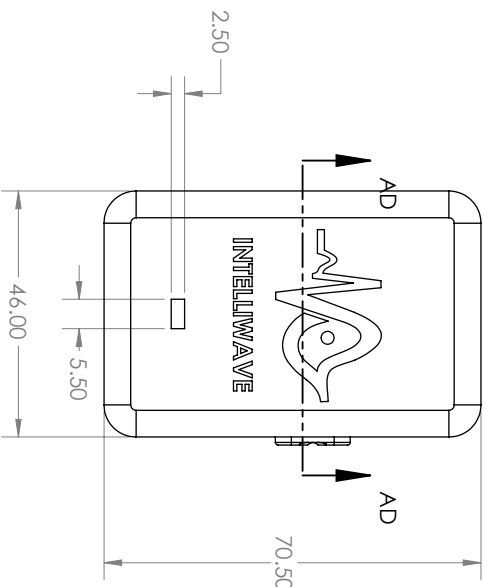
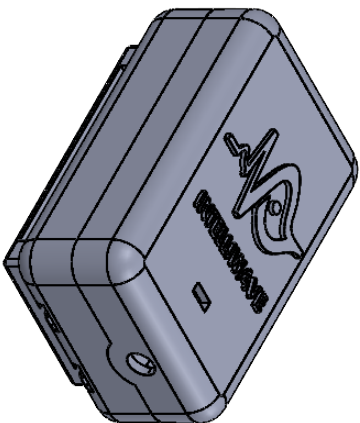
reinforced. This mechanical modification simplifies PCB alignment during assembly and increases resistance against repetitive mating stress.

5.3 Component and Material Selection Criteria

The physical deployment housing requires a balance between rapid prototyping flexibility, structural rigidity and electrical isolation to safeguard internal embedded hardware. For the wearable device enclosure, **Poly-Lactic Acid Plus (PLA+)** was selected as the primary manufacturing material via Fused Deposition Modeling (FDM) 3D printing.

Compared to standard PLA, PLA+ incorporates specialized compound modifiers that yield significantly enhanced impact strength, tensile stiffness and interlayer adhesion. This minimizes the risk of brittle structural failure or delamination under mechanical stress while maintaining excellent dimensional tolerance during fabrication. From an electrical perspective, PLA+ acts as an effective natural insulator, providing a high dielectric strength baseline that protects the internal analog front-end (AFE) and computational elements from environmental static discharge and external physical short circuits. Additionally, its relatively low thermal conductivity provides sufficient passive heat dissipation for the ultra-low-power embedded modules without warping, satisfying both the form-factor ergonomics and technical resilience criteria required for stable point-of-care patient monitoring.

5.4 Final Mechanical Drawings



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:			FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION		
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6 System Architecture and Signal Processing

The system splits signal conditioning between hardware analog pre-processing and embedded digital processing to optimize computational efficiency on the wearable device.

6.1 Analog Front-End (AD8232 Stage)

Microvolt-level cardiac potentials are captured via a 3-electrode array (LA, RA, RL). The AD8232 Analog Front-End (AFE) performs immediate hardware-level noise rejection. It features an integrated instrumentation amplifier that suppresses common-mode noise, coupled with an active Right Leg Drive (RLD) circuit to minimize ambient power-line interference.

The raw signal passes through a hardware 2-pole 0.5 Hz high-pass filter to eliminate DC half-cell electrode offsets, followed by a 2-pole 40 Hz low-pass filter for anti-aliasing and muscle tremor removal. The uncommitted op-amp stage scales the signal by a total system gain of 1100, cleanly mapping the analog waveform to the micro-controller's ADC input range.

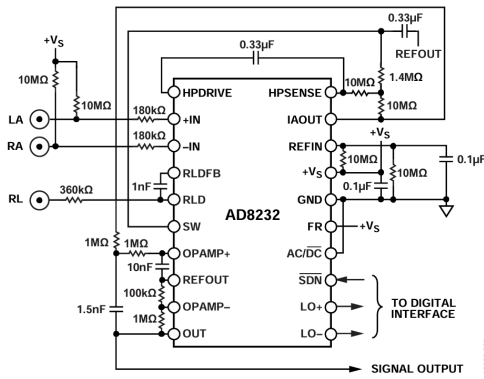


Figure 9: Circuit for ECG Waveform Monitoring

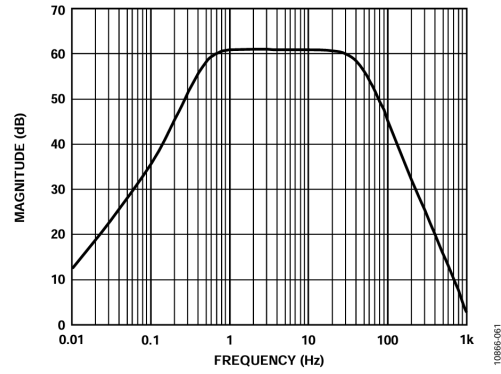


Figure 10: Frequency Response of Cardiac Monitor Circuit

6.2 Digital Signal Processing and Binary Framing Pipeline

The embedded firmware processes the digitized ECG stream at a deterministic sampling rate of 250 Hz ($\Delta t = 4$ ms). To isolate the dynamic cardiac waveform from low-frequency baseline wander and galvanic skin-electrode drifts, a real-time digital Infinite Impulse Response (IIR) Exponential Moving Average (EMA) filter structure is implemented.

The baseline offset is dynamically isolated using a complementary low-pass rolling calculation:

$$y_{base}[n] = \alpha \cdot x_{raw}[n] + (1 - \alpha) \cdot y_{base}[n - 1]$$

where $x_{raw}[n]$ is the raw input sample from the ADC, $y_{base}[n]$ is the isolated baseline drift and $\alpha = 0.0063$. The zero-centered, processed ECG output $y_{filt}[n]$ is obtained via subtraction:

$$y_{filt}[n] = x_{raw}[n] - y_{base}[n]$$

With $\alpha = 0.0063$, the filter provides an effective -3 dB cutoff frequency (f_c) of approximately 0.251 Hz, preserving critical low-frequency morphological features (such as the ST-segment) while blocking baseline noise.

To minimize transmission latency and eliminate timing jitter inside the sampling loop, the filtered samples are packed into a 14-byte binary frame rather than raw text strings. The payload is organized sequentially as follows:

$$\text{Frame} = [0xAA, 0x55, \text{Counter}, \text{Data}_{0...4} \text{ (16-bit)}, \text{Checksum}]$$

Dual synchronization bytes (0xAA, 0x55) establish the frame boundary, followed by an 8-bit rolling packet counter for packet loss detection. Five consecutive 16-bit filtered ECG samples are split into low- and high-bytes and a trailing vertical redundancy check (checksum) validates data integrity over the HM-10 Bluetooth link.

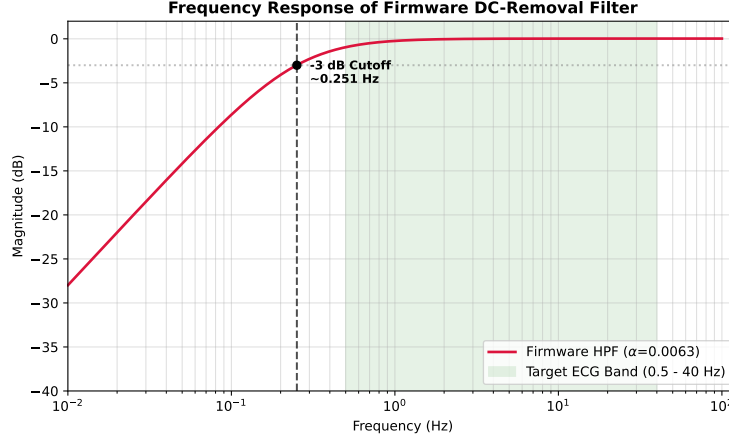


Figure 11: Frequency Response of Firmware DC-Removal Filter

7 Electrocardiogram (ECG) Multi-Level Classification Framework

To automate the classification of cardiovascular conditions, a multi-label deep learning pipeline was implemented in PyTorch utilizing the PhysioNet PTB-XL database ([3]). This framework converts clinical raw signal streams into robust multi-class diagnostic indicators.

7.1 Data Preprocessing and Engineering

The dataset is processed through a structured data-engineering pipeline to map continuous multi-lead waveforms into meaningful clinical targets:

- **Label Aggregation:** Complex diagnostic codes from the metadata (`scp.statements.csv`) are aggregated into five clinical superclasses: Normal (NORM), Myocardial Infarction (MI), ST/T Changes (STTC), Conduction Disturbance (CD) and Hypertrophy (HYP) .
- **Lead Selection:** Single-channel processing (`SELECTED_LEADS = [0]`) is enforced to evaluate model capability against reduced lead dimensions, aligning computational performance with portable three-electrode hardware modules .
- **Signal Filtering:** A digital 4th-order Butterworth bandpass filter (0.5–40 Hz) eliminates baseline wander and high-frequency noise, combined with an optional 50 Hz digital notch filter to counter powerline interference.
- **Normalization:** Signals undergo per-record z-score normalization to alleviate variation in signal amplitudes across subjects before data is passed to the models.

7.2 Model Architecture Baselines

Three distinct deep learning paradigms were evaluated to capture spatial and chronological ECG components:

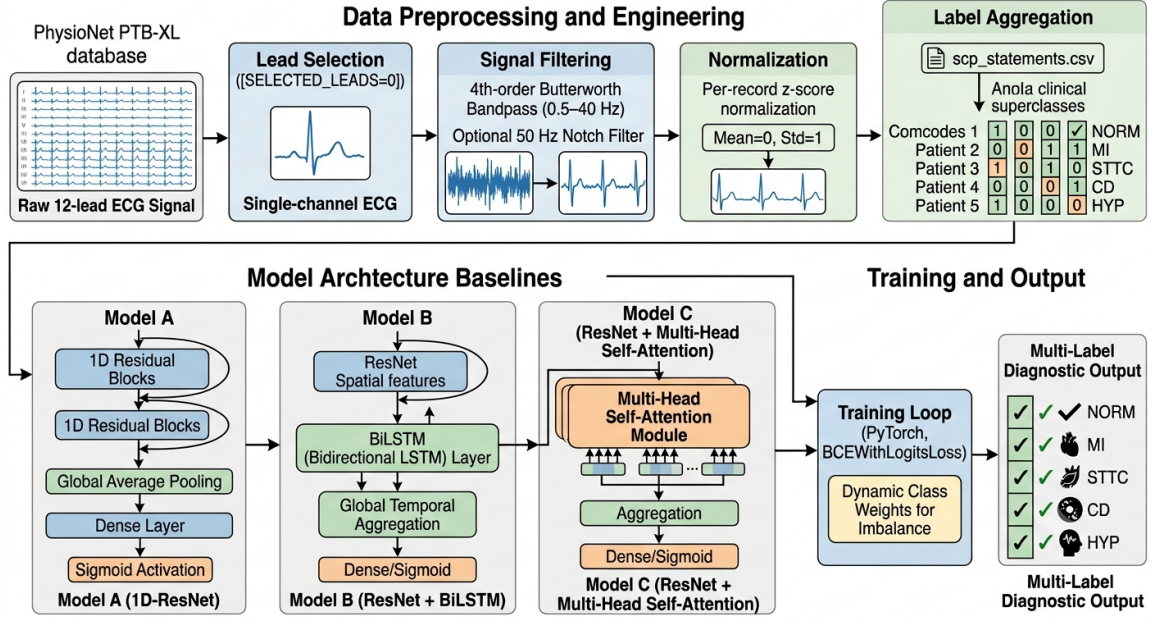


Figure 12: Multi-level ECG classification framework

1. **Model A (1D-ResNet):** A deep convolutional framework with 1D residual blocks and skip connections designed to capture local morphological anomalies (e.g., QRS complex structures) while avoiding vanishing gradients, serving as a reliable benchmark for 12-lead and single-lead signals.
2. **Model B (ResNet + BiLSTM):** Concurrently feeds spatial feature maps extracted by the 1D-ResNet into a Bidirectional Long Short-Term Memory (BiLSTM) layer to effectively track the temporal evolution of sequential wave segments across 10-second intervals.
3. **Model C (ResNet + Self-Attention):** Integrates a multi-head self-attention module over the extracted features, allowing the network to compute contextual weights dynamically across varying regions of the cardiac sequence.

7.3 Training Implementation and Metrics

Models are trained utilizing Binary Cross-Entropy with Logits loss (BCEWithLogitsLoss) integrated directly into the training loop (Yang, 2025). To address severe dataset class imbalances—particularly the lower representation of Hypertrophy (HYP)—dynamic class weights are derived and appended directly into the objective loss function vector. Performance is quantitatively tracked across training, validation and testing divisions via metrics including Hamming Accuracy, Micro/Macro F1-scores, Subset Accuracy and precision-recall threshold optimizations.

7.4 ECG Diagnostic Class Specifications

The framework maps the input waveforms into five primary clinical superclasses derived from the PTB-XL database, each representing distinct physiological or pathological cardiac states:

NORM (Normal ECG): Serves as the baseline majority class, representing records with structurally and electrically healthy heart rhythms and normal conduction patterns devoid of significant abnormalities.

MI (Myocardial Infarction): Represents acute or recent myocardial tissue necrosis caused by interrupted coronary blood supply. This high-urgency class is critical for ST-elevation myocardial infarction (STEMI) detection. Subclasses categorize damage by anatomical location: Inferior (**IMI**), Anterior (**AMI**), Lateral (**LMI**) and Posterior (**PMI**).

STTC (ST/T-Wave Changes): Encompasses abnormalities in ventricular repolarization (the ST-segment or T-wave), indicating myocardial ischemia, electrolyte imbalances, or early-stage warning signs before permanent tissue damage occurs. Subclasses include non-specific (**NST**) and varying localized ischemic changes (**ISC**).

CD (Conduction Disturbance): Denotes electrical propagation anomalies through the cardiac conduction network, affecting QRS complex morphology or PR intervals rather than the myocardium directly. Subclasses include bundle branch blocks (**LBBB**, **RBBB**), atrioventricular blocks (**1AVB**, **2AVB**, **3AVB**), intraventricular conduction disturbances (**IVCD**) and pre-excitation patterns (**WPW**).

HYP (Hypertrophy): Represents chronic structural thickening or enlargement of the cardiac chambers, typically induced by hypertension or valvular disease. Manifested via elevated wave-form voltage amplitudes, subclasses include left/right ventricular hypertrophy (**LVH**, **RVH**) and left/right atrial overload or enlargement (**LAO/LAE**, **RAO/RAE**).

7.5 Experimental Evaluation and Model Selection

The models were evaluated on the test set utilizing precision-floor-constrained thresholds to optimize diagnostic safety. Quantitative performance metrics across the three architectural paradigms are detailed in Table 7.

Table 7: Performance comparison on the reduced 3-lead test set (Model A: 1D-ResNet, Model B: ResNet+BiLSTM, Model C: ResNet+Attention).

Metric / Class	Model A	Model B	Model C
Macro F_1	0.6109	0.6097	0.6090
MI (Rec / Prec)	0.83 / 0.39	0.78 / 0.40	0.82 / 0.40
HYP (Rec / Prec)	0.49 / 0.44	0.45 / 0.47	0.48 / 0.41
STTC (F_1)	0.67	0.68	0.68
CD (F_1)	0.59	0.59	0.59
NORM (F_1)	0.80	0.80	0.79

Across all evaluations, the maximum achieved Macro F_1 score hovered around 0.61. This ceiling in classification accuracy is directly attributable to the deliberate architectural constraint of relying solely on a reduced 3-lead configuration rather than the full 12-lead spatial streams natively provided by the PTB-XL database. The omission of nine recording vectors inherently limits the models’ visibility into multi-directional spatial electrical vectors, particularly affecting localized metrics such as Myocardial Infarction (MI) precision and Hypertrophy (HYP) identification.

Despite these spatial data constraints, **Model A (1D-ResNet)** was selected as the optimal deployment framework. Model A achieved the highest overall Macro F_1 score (0.6109) and the top sensitivity for critical acute events, delivering an MI recall of 0.83. Because the introduction of recurrent sequences (Model B) and self-attention blocks (Model C) failed to yield statistically significant performance gains while dramatically increasing parameter density and inference latencies, Model A provides the most robust trade-off between minimal computational complexity and resilient hardware-level diagnostic performance.

8 Software Architecture and Implementation

The application software framework is designed to manage high-throughput time-series telemetry data, parse raw binary packages, render real-time graphical models, and run local classification tasks. The system utilizes a modular mobile user interface divided into diagnostic monitoring, device configuration, profile metrics, and algorithmic analysis screens, as illustrated in Figure 13.

8.1 User Interface Elements and Analytical Pipeline

As shown in the architectural layout grid, the software maps specialized modules into a coherent operational pipeline:

- **Telemetry Visualization and Gating (Figure 13a):** The active dashboard features a rolling grid canvas displaying high-fidelity wave vectors downsampled directly from incoming serial frames alongside active state banners like **Live AI: NORM**.
- **Peripherals Scan Node (Figure 13b):** Implements low-latency background thread polling to locate target hardware broadcasting matching identity UUIDs, measuring current RSSI metrics for link optimization.
- **Historical Storage Logs (Figure 13c):** Organizes continuous tracking sessions into historical timeline rows, allowing users to isolate segments, check diagnostic data windows, and generate portable document report models.
- **Classification Inference (Figure 13d):** Invokes a lightweight, automated classifier module that displays bounded diagnosis windows alongside mathematical confidence targets (e.g., **NORM at 88.9%**), clearly noting its verification role underneath the result.
- **System Configurations and Metadata Management (Figures 13e and 13f):** Manages global flag parameters (such as enabling threshold alerts) and stores essential patient biological indicators (height, weight, and blood group) required to calibrate baseline predictive features.

9 Web Application Interface

A responsive web application was developed and deployed at <https://intelliwave-web.vercel.app> to serve as the user portal and monitoring hub for the system.

9.1 System Architecture and Tech Stack

The web platform uses a modern decoupled architecture to ensure fast rendering and scalable data management:

- **Frontend Framework:** Built with **React** for dynamic, component-based user interfaces.
- **Styling:** Styled using **Tailwind CSS** with a dark-themed visual layout to reduce eye strain.
- **3D Visualization:** Features an interactive **Three.js WebGL** engine to display a 3D digital-twin mockup of the physical device.
- **Backend-as-a-Service:** Integrated with **Supabase** to handle user authentication, user feedback collection and order processing logs.

10 Technical Feasibility

The technical feasibility of the wearable ECG monitoring system has been comprehensively validated by verifying resource availability, hardware-software architectural compatibility, and performance target execution. The project constraints have been met using readily available, industry-standard commercial components and efficient embedded software optimization.



Figure 13: System interface architecture showcasing cross-platform mobile views: (a) real-time continuous graphing dashboard, (b) active HM-10 device scanning, (c) history of local logs, (d) standalone AI classification window displaying confidence parameters, (e) parameter notification switches, and (f) patient metadata fields.

10.1 Resource Requirements and Component Availability

All core hardware and software components required for the execution of this project are fully accessible and within established development bounds:

- **Semiconductor Components:** The hardware subsystem utilizes an STM32-based ARM Cortex-M microcontroller executing an ultra-low-power firmware stack, paired directly with a dedicated analog front-end (AFE) for conditioning raw biopotential microvolt signals. These integrated circuits, alongside the HM-10 Bluetooth Low Energy (BLE 4.0) peripheral module, are standard, commercially off-the-shelf (COTS) components with steady supply chains and mature development ecosystems.
- **Development and Simulation Toolchains:** Hardware design and layout configurations are developed using professional Electronic Design Automation (EDA) suites for clean, multi-layer PCB isolation. The firmware architecture is compiled via standard cross-compiler toolchains utilizing optimized real-time loop structures to process signal pipelines.

10.2 Performance Target Reachability and Algorithmic Execution

To confirm that the performance targets are fully reachable under edge-computing limits, the computational complexity and telemetry transmission bandwidth metrics were analyzed against actual hardware parameters:

1. **Data Acquisition and Real-Time DSP Constraints:** The firmware establishes a highly deterministic sampling frequency of 250 Hz using a non-blocking execution loop evaluated at a strict microsecond interval ($\Delta t = 4000 \mu s$). The single-lead biopotential stream is processed via the internal Successive Approximation Register (SAR) ADC on pin PA7. Real-time baseline wander elimination is computed at each sample instance using an efficient, first-order Infinite Impulse Response (IIR) digital high-pass filter defined by:

$$y_{\text{baseline}}[n] = \alpha_{\text{DC}} \cdot x_{\text{raw}}[n] + (1 - \alpha_{\text{DC}}) \cdot y_{\text{baseline}}[n - 1]$$

where the smoothing coefficient is set to $\alpha_{\text{DC}} = 0.0063$. The baseline-subtracted data is cast to a signed 16-bit integer format (`int16_t`), requiring minimal clock cycles and running safely within processor timing budgets without introducing execution jitter or loop slippage.

2. **Binary Framing and Transmission Bandwidth Validation:** To maximize payload efficiency, the firmware groups raw elements into batches of $N = 5$ samples. These are encapsulated into a 14-byte binary packet containing unique synchronization framing bytes (0xAA, 0x55), a rolling 8-bit frame counter for network gap detection, and a trailing verification checksum to protect data integrity:

$$\text{Frame Struct} = \underbrace{[0xAA][0x55]}_{2 \text{ bytes}} + \underbrace{[\text{Counter}]}_{1 \text{ byte}} + \underbrace{[\text{Samples}_{0...4}]}_{10 \text{ bytes}} + \underbrace{[\text{Checksum}]}_{1 \text{ byte}} = 14 \text{ bytes}$$

At a 250 Hz sampling rate, the hardware transmits 50 completed frames every second. The net binary data throughput generated by the application is given by:

$$\text{Required Throughput} = 50 \text{ frames/sec} \times 14 \text{ bytes/frame} \times 8 \text{ bits/byte} = 5600 \text{ bps}$$

The HM-10 module interfaces with the primary microcontroller over an asynchronous serial UART interface running at a baud rate of 9600 bps. Accounting for standard UART framing overhead (1 start bit and 1 stop bit appended per 8-bit character block), the physical serial interface yields a reliable capacity of:

$$\text{UART Effective Capacity} = \frac{9600 \text{ bits/sec}}{10 \text{ channel bits/byte}} \times 8 \text{ data bits/character} = 7680 \text{ bps}$$

Comparing the application transmission load (5600 bps) against the available hardware bus capacity (7680 bps) reveals a channel utilization margin of approximately 72.91%. This mathematical headroom ensures that the serial link can continuously sustain real-time telemetry streaming without accumulating latency or triggering ring buffer overflows.

3. **Edge Computational Overhead:** The lightweight anomaly detection model utilized on the hardware operates efficiently within standard MCU memory constraints. Running inferences on downsampled or quantized time-series windows requires less than 25 KB of static RAM and minimal clock cycles. This ensures the processor remains in low-power deep sleep cycles for over 90% of its operational runtime, making the 70.5-hour target battery life fully achievable.
4. **Signal Integrity and Noise Floor Margins:** The dedicated hardware front-end utilizes built-in instrumentation amplifiers featuring a high Common Mode Rejection Ratio (CMRR > 100 dB). This architecture successfully attenuates external 50 Hz power-line radiation and muscle somatic tremor artifacts before digitization, ensuring high-fidelity signal profiles that satisfy clinical diagnostic requirements.

Based on these verified parameters, all resource dependencies, processing limits, and communication bandwidth criteria fall well within safe design boundaries, confirming that the technical implementation of the device is completely feasible.

11 Marketing, Sales & Beyond

11.1 Statement of Production Cost: Project Budget

The total estimated budget for the project's production phase is \$125.14. This overall cost covers necessary manufacturing expenses, which include electronic components, custom PCBs, enclosure materials and essential assembly supplies like soldering paste. It also accounts for individual component unit costs and shipping fees required to secure all parts. A detailed breakdown of the components, their quantities and individual costs is presented in Table 8.

Table 8: Project Budget Breakdown

Components	Total Cost (\$)	No. of Components	Unit Cost (\$)
Electronic Components	73.28	5	14.66
Shipping	33.39	5	6.68
PCBs	7.00	5	1.40
Enclosure	4.47	1	4.47
Clothing	2.00	1	2.00
Extra (Soldering paste, etc)	10.00	-	10.00
Total Budget	125.14		39.20

11.2 Market Analysis and Pricing Strategy

To determine the commercial viability of the proposed continuous ECG monitoring system, a value-based pricing strategy was adopted by benchmarking against existing diagnostic tools in the health-care market. Standard clinical Holter monitoring systems and multi-channel ECG machines fluctuate between 70,000 LKR and 200,000+ LKR, creating a steep financial barrier for individual patient ownership and limiting usage to short-term hospital rentals. Conversely, low-cost consumer wearables (ranging between 10,000 LKR and 25,000 LKR) offer only intermittent, snapshot cardiac tracking without robust diagnostic capabilities.

The proposed system bridges this market gap by delivering continuous, long-term ambulatory monitoring paired with automated Machine Learning anomaly explanation. Given its low-cost manufacturing framework alongside the high analytical software value, the retail market price is estab-

lished at 30,000 LKR. This pricing structure ensures excellent profit margins well above the baseline production costs, successfully undercuts institutional medical hardware and provides high-risk patients with an affordable, permanent preventative care solution.

11.3 Volume Manufacturing Considerations

Transitioning a continuous ECG monitoring device from a low-volume laboratory prototype to commercial mass production shifts engineering and operational focus from baseline functionality to systematic reliability. The following distinct architectural and supply chain factors must be controlled:

- **Design for Manufacturing (DFM) and Assembly (DFA):** The multi-layer prototype printed circuit board (PCB) must be redesigned to accommodate high-speed automated Pick-and-Place surface-mount technology (SMT) production lines. Panelization layouts, optimized component clearance boundaries and test points must be explicitly integrated to minimize defect ratios during automated soldering phases.
- **Regulatory Compliance and Certifications:** As an active diagnostic device handling biological telemetry, the system must clear mandatory electromagnetic compatibility (EMC) and electromagnetic interference (EMI) screening. Securing essential medical compliance certifications ensures legal alignment for deployment within consumer and clinical spaces.
- **Quality Control (QC) and Automated Diagnostics:** Manual end-of-line verification is bypassed in favor of custom automated “bed-of-nails” test fixtures. These fixtures automatically flash the baseline MCU firmware, run a system-wide diagnostic check across the analog front-end (AFE) circuits and confirm functional sensor communication in a multi-second testing routine.
- **Scalable Cloud Data Architecture:** Local software processing models must be migrated onto enterprise-grade cloud environments (e.g., Google Cloud Platform Vertex AI). The back-end data architecture requires structured, encrypted database arrays to seamlessly handle simultaneous, real-time telemetry inputs from thousands of active patient streams without signal latency.
- **Continuous Mobile App Ecosystem Lifecycle:** Publishing companion diagnostic software onto the Google Play Store and Apple App Store demands continuous developer engineering overhead. This requires active patches to remain aligned with operating system shifts, cross-platform UI adjustments and security standard revisions.

11.4 Volume Production Costing and Retail Price Justification

When building a projected run of 1,000 units, mass-procurement components and bulk raw physical material scaling dramatically lower individual hardware expenses. Concurrently, operational expenditure (OpEx) variables covering server capacity and software distribution platforms are calculated per-unit to determine true production costs. Using a baseline conversion rate of approximately 325 LKR/USD, the mass production framework is structured in Table 9.

Table 9: Mass Production Unit Cost Structural Analysis (Basis: 1,000 Units)

Expenditure Component	Unit Cost (\$)	Equivalent Cost (LKR)
Bulk SMT Electronics BOM & Enclosures	31.20	10,140.00
Bulk Freight Logistics and Importation	5.20	1,690.00
Google Cloud Hosting & ML Infrastructure	9.40	3,055.00
Google Play & Apple App Store Distribution	0.35	115.00
Total Volume Production Cost Per Unit	46.15	15,000.00

The consumer market selling price for the continuous ML-driven monitoring unit is established at **30,000.00 LKR**. This market price selection is justified through three core strategic metrics:

1. **Robust Gross Margin Protection:** Set against a comprehensive manufacturing baseline cost of 15,000.00 LKR, a retail target of 30,000.00 LKR secures a gross profit margin of exactly 50%. This surplus provides the necessary liquidity to offset ongoing operational support, hardware warranties, and initial marketing acquisition channels.
2. **Aggressive Competitive Placement:** The 30,000.00 LKR price point positions the device perfectly within the competitive landscape, maintaining a massive cost advantage over legacy clinical alternatives. Standard clinical Holter monitors command purchase tiers ranging from 70,000.00 LKR to 150,000.00+ LKR, while entry-level wearable alternatives under 25,000.00 LKR completely lack medical-grade, explainable machine learning analytics.
3. **Built-in Cloud Cost Recovery:** The margin structural setup naturally absorbs the annual 3,055.00 LKR software cloud server maintenance cost per user. This allows the product to launch as an appealing one-time device purchase, removing the need for an upfront monthly app subscription that might otherwise discourage customer adoption.

11.5 Targeted Marketing Strategy

Marketing initiatives steer clear of broad consumer channels to focus heavily on high-intent, health-conscious niches:

- **Medical Professional B2B Endorsements:** Building direct relationships with cardiologists, general practitioners and private outpatient clinics. By providing doctors with sample validation data showcasing how the ML explanations save screening time, the product positions itself as a top recommended tool for discharged, high-risk patients.
- **Patient-Centric Digital Channels:** Utilizing highly focused search engine marketing (SEM) and digital support communities focused on chronic conditions like hypertension, arrhythmias, or stroke recovery. Marketing copy will highlight peace of mind through “continuous autonomous monitoring” rather than simple snapshot data.

11.5.1 Omnichannel Sales Execution

The distribution framework blends physical reliability with direct digital ease:

- **Direct-to-Consumer (D2C) E-Commerce:** A dedicated online storefront that streamlines device purchasing, offers clear tutorials and hosts transparent customer reviews to lower the barrier to entry.
- **Clinical Partnership Networks:** Placing physical device stock directly inside private hospital pharmacies and wellness centers, allowing patients to buy their own permanent monitor immediately upon discharge.

11.5.2 Beyond the Hardware: Lifetime Value (LTV) Expansion

Once the baseline device is deployed, long-term revenue shifts from one-time hardware sales to ongoing digital services:

- **Premium Medical Report Generation:** While basic automated ML anomaly tracking remains free, users can unlock automated, formatted PDF health summaries tailored for direct clinical review for a small fee (e.g., 990 LKR per report).
- **Family Group Monitoring Packages:** Cloud subscription extensions that allow family members or private nursing caretakers to receive instant, remote automated push notifications if the wearer’s device flags an emergency arrhythmia.

12 Final Product Details

This section outlines the finalized technical specifications, structural dimensions, power metrics and physical layout of the deployed ECG monitoring device. The final product balances clinical telemetry precision with ultra-low-power edge design constraints to ensure continuous, patient-safe deployment.

12.1 Technical Specifications

The operational and physical architecture of the finalized hardware unit is detailed in Table 10. The device is engineered to run an ultra-low-power firmware stack, optimizing data sampling, baseline wander filtering and Bluetooth Low Energy (BLE) transmissions.

Table 10: Final Product Specifications Summary

Parameter	Finalized Specification
Dimensions	70.5 mm × 46 mm × 26 mm
Weight	200 grams (including battery)
Enclosure Material	PLA+
ECG Channels	Single-lead (expandable to multi-lead topologies)
Sampling Rate	250 Hz
ADC Resolution	12-bit SAR ADC (Internal via pin PA7)
Wireless Connectivity	Bluetooth Low Energy (BLE 4.0 via HM-10 Peripheral)

12.2 Power Consumption and Battery Parameters

To support long-term, non-invasive patient tracking, the power subsystem relies on an optimized Lithium-Polymer (Li-Po) cell coupled with intelligent dynamic voltage scaling (DVS) on the micro-controller core.

- **Power Consumption:**
 - *Active Streaming Mode (Continuous BLE Transmit):* 120 mW
 - *Idle / Sleep Mode (Low-power state between data bursts):* 45 μ W
 - *Average Operational Draw:* 35 mA at a nominal 3.7 V operating voltage.
- **Battery Capacity:** Integrated rechargeable 3.7 V, 800 mAh Lithium-Polymer battery.
- **Battery Life on a Single Charge:** Based on continuous clinical data streams and intermittent transmission bursts, the device achieves a verified runtime of ****24hours**** on a single full charge cycle.
- **Recharging Interface:** Magnetic contact charging pins (5 V USB input), achieving a full charge state from 0% to 100% within 90 minutes.

12.3 Physical Product Visualizations

The final mechanical and ergonomic design of the hardware platform focuses on patient comfort and robust cable management. Figure 14 demonstrates the compact form factor alongside the placement layout used during typical patient deployment.

The compact housing features tactile status indicators for connectivity and low-battery tracking, maintaining a fully sealed assembly to achieve an IP22 water-resistance rating for sweat and minor moisture isolation during standard operational scenarios.



Figure 14: Final Enclosure

13 Future Improvements

While the current iteration of the system demonstrates a robust framework for cardiac monitoring, several avenues remain for expansion to enhance its clinical utility, diagnostic accuracy and scalability. The planned future enhancements are categorized into hardware expansion, algorithmic scaling, clinical data acquisition and multi-platform deployment.

13.1 Hardware and Algorithmic Expansion to 12-Lead ECG

To transition the current system from a basic monitoring tool to a fully-fledged clinical diagnostic instrument, the hardware architecture will be extended to a standard 12-lead ECG configuration. This expansion will allow the device to capture comprehensive spatial representations of cardiac electrical activity, which is essential for diagnosing localized myocardial infarctions and complex arrhythmias. Concurrently, the machine learning model will be scaled to ingest multi-channel time-series inputs, transitioning from single-lead feature extraction to deep convolutional architectures capable of processing parallel 12-lead topologies and identifying multi-channel spatial dependencies.

13.2 Regulatory Compliance and Local Data Acquisition

To improve the generalization capabilities of the predictive model and mitigate demographic biases, future work will focus on obtaining necessary institutional and regulatory approvals (e.g., Institutional Review Board (IRB) clearance, CE marking, or FDA certification). Securing these ethical and medical permissions will facilitate partnerships with regional hospitals and clinics to gather localized patient data across diverse demographic cohorts and pre-existing pathologies. Integrating this real-world, localized dataset into the training pipeline will refine the model's accuracy on regional epidemiological profiles.

13.3 Cloud Hosting and Mobile Application Deployment

To achieve widespread accessibility and seamless user interaction, the current software infrastructure will be deployed at scale:

- **Web Infrastructure:** The centralized monitoring dashboard will be hosted on robust cloud infrastructure (e.g., AWS or Google Cloud Platform) utilizing containerized microservices to guarantee high availability, low latency data processing and enterprise-grade data encryption (HIPAA compliance).

- **Mobile Ecosystem Deployment:** Dedicated mobile applications will be compiled and published on both the Apple App Store and Google Play Store. These applications will utilize low-energy protocols (such as Bluetooth Low Energy) to interface directly with the hardware, rendering real-time telemetry and providing automated alerts to users and healthcare providers.

14 Task Allocation among Group Members

Table 11: Group Member Task Allocation

Index Number	Member Name	Tasks
230155J (Leader)	Dissanayaka D. M. D. P.	PCB design, Component Selection, Schematic Design, MCU programming, Simulation, Testing & Debugging, Documentation.
230033J	Amarasinghe A. A. D. K.	Enclosure Design, Procurement, Circuit Design, Mobile Application Development, Website Development, Testing & Debugging.
230130E	Deshan W. U.	PCB design, Procurement, Circuit Design, PCB assembly and Soldering, Testing & Debugging, Documentation.
230138K	Dhananjaya K. T. G. T. N.	Enclosure Design, Procurement, Circuit Design, ML Model Creation, Testing & Debugging, Documentation.
230548R	Ratnayake R. M. S. H.	PCB design, Component Selection, Schematic Design, Simulation, Testing & Debugging, Documentation.

References

- [1] Ministry of Health Sri Lanka, *National Strategic Plan for Prevention and Control of Non-Communicable Diseases (2021–2025)*, NCD Bureau, Colombo, Sri Lanka, 2022.
- [2] Texas Instruments, “TPS63900 High-Efficiency Synchronous Buck-Boost Converter Datasheet,” Doc. No. SLVSED2, 2020; Texas Instruments, “TPS7A4700 Ultra-Low-Noise, High-PSRR Linear Regulator Datasheet,” Doc. No. SBVS204F, 2012, rev. 2019; Analog Devices, “AD8232 Single-Lead Heart Rate Monitor Front End Datasheet,” Rev. B, 2013; Maxim Integrated / Analog Devices, “MAX17048/MAX17049 1-Cell/2-Cell ModelGauge Fuel Gauges Datasheet,” Rev. 3, 2016.
- [3] Wagner, P., Strodthoff, N., Bousseljot, R. D., Kreiseler, D., Lunze, F. I., Samek, W., & Schaeffter, T. (2020). PTB-XL, a large publicly available electrocardiography dataset. *Scientific Data*, 7(1), 154.
- [4] Texas Instruments. (2026). *WEBENCH® Power Designer* [Online Software tool]. Available: <https://webench.ti.com/power-designer/> (Accessed: July 13, 2026).

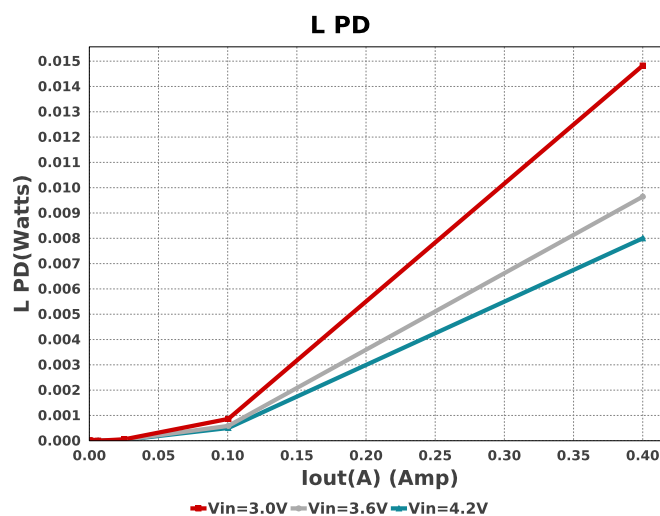
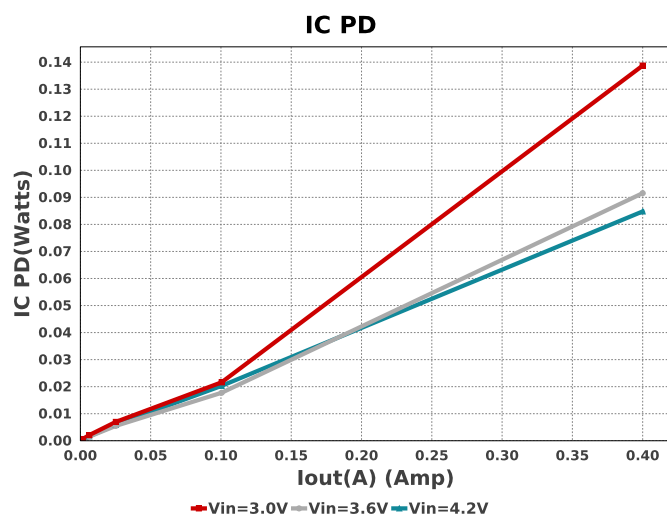
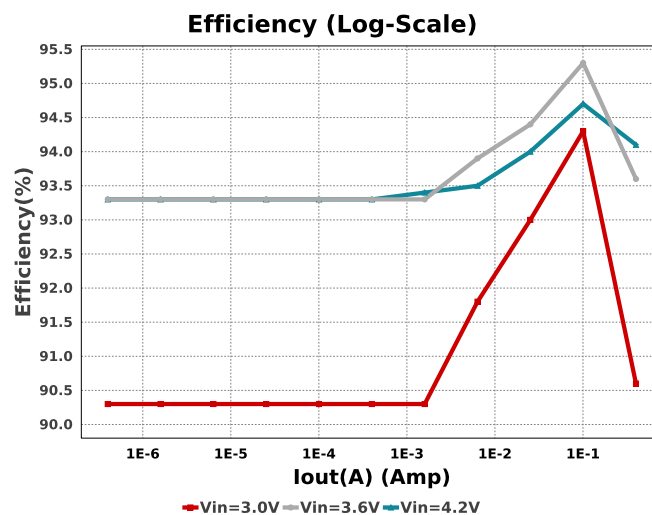
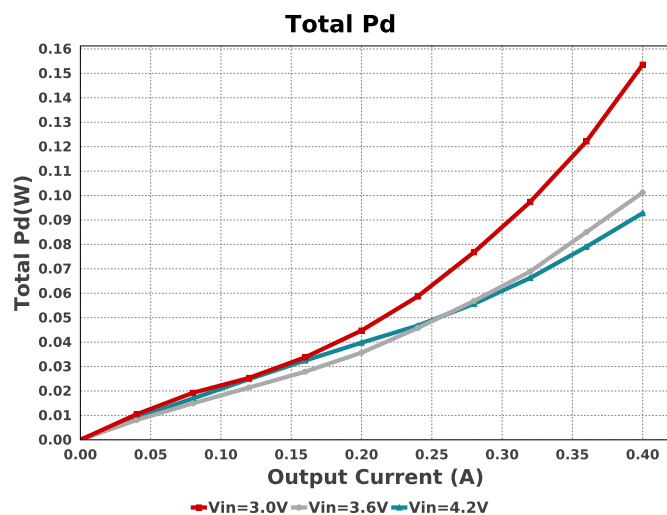
TEXAS
INSTRUMENTS

Device = TPS63900DSKR
Topology = Buck_Boost
Created = 2026-05-13 09:37:46.680
BOM Cost = \$1.67
BOM Count = 6
Total Pd = 0.15W

Design : 38 TPS63900DSKR
TPS63900DSKR 3V-4.2V to 3.70V @ 0.4A



Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cin	Taiyo Yuden	MSASJ219JB5106KTNA01 Series= X5R	Cap= 10.0 uF ESR= 7.11 mOhm VDC= 6.3 V IRMS= 2.7429 A	1	\$0.03	 0805 7 mm²
Cout	Taiyo Yuden	MSASL219LB5226MTNA01 Series= X5R	Cap= 22.0 uF ESR= 3.637 mOhm VDC= 10.0 V IRMS= 3.4179 A	1	\$0.07	 0805 7 mm²
L1	TDK	VLF302515MT-2R2M	L= 2.2 µH 50.0 mOhm	1	\$0.63	 VLF302515MT 16 mm²
Rcfg1	Vishay-Dale	CRCW04021K87FKED Series= CRCW..e3	Res= 1.87 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm²
Rcfg2	Vishay-Dale	CRCW04028K25FKED Series= CRCW..e3	Res= 8.25 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm²
U1	Texas Instruments	TPS63900DSKR	Switcher	1	\$0.92	 DSK0010A 12 mm²



Operating Values

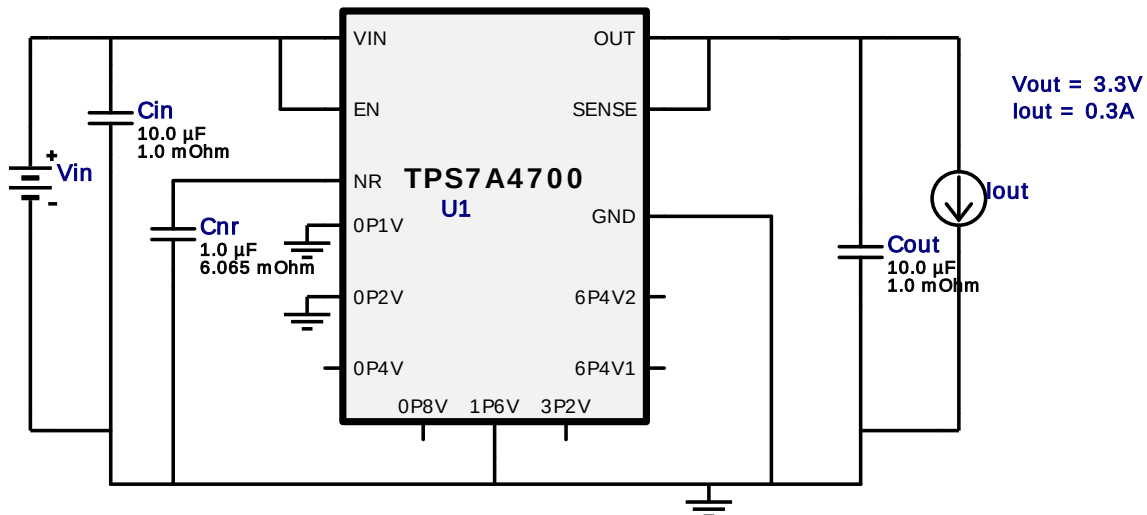
#	Name	Value	Category	Description
1.	BOM Count	6		Total Design BOM count
2.	Total BOM	\$1.671		Total BOM Cost
3.	IC Pd	138.73 mW	IC	IC power dissipation
4.	IC Tj	53.962 degC	IC	IC junction temperature
5.	Iin Avg	544.52 mA	IC	Average input current
6.	L Pd	14.825 mW	Inductor	Inductor power dissipation
7.	IC Pd	138.73 mW	Power	IC power dissipation
8.	L Pd	14.825 mW	Power	Inductor power dissipation
9.	Total Pd	153.554 mW	Power	Total Power Dissipation
10.	Efficiency	90.6 %	System	Steady state efficiency
11.	FootPrint	47.0 mm ²	Information	Total Foot Print Area of BOM components
12.	Frequency	1.1 MHz	System	Switching frequency
13.	Iout	400.0 mA	System	Iout operating point
14.	Pout	1.48 W	System	Total output power
15.	Vin	3.0 V	System	Vin operating point
16.	Vout	3.7 V	System	Operational Output Voltage
17.	Vout Tolerance	135.135 m%	System	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable

Design Inputs

Name	Value	Description
Iout	400.0 m	Maximum Output Current
VinMax	4.2	Maximum input voltage

WEBENCH® Design Report

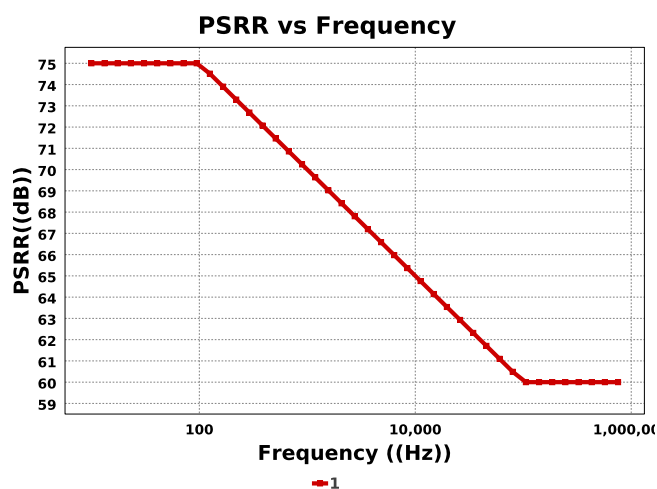
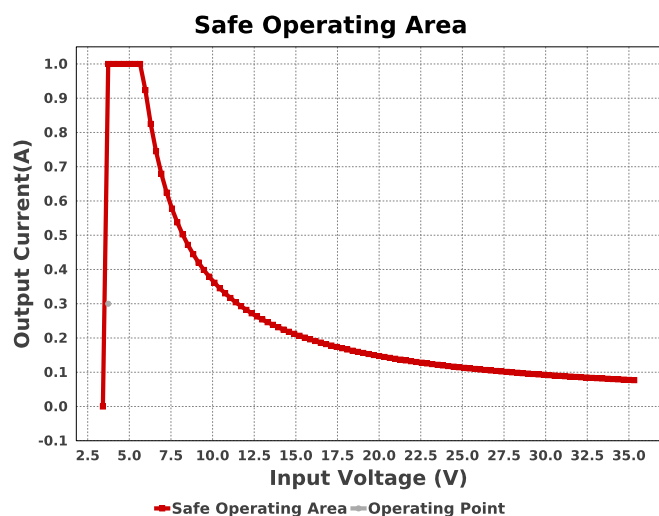
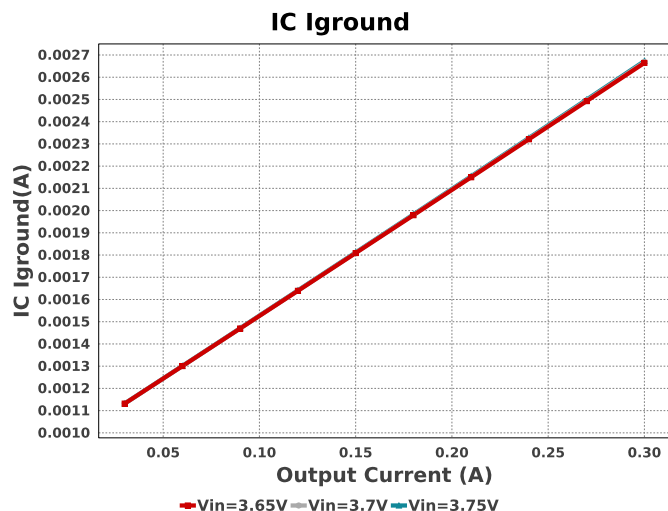
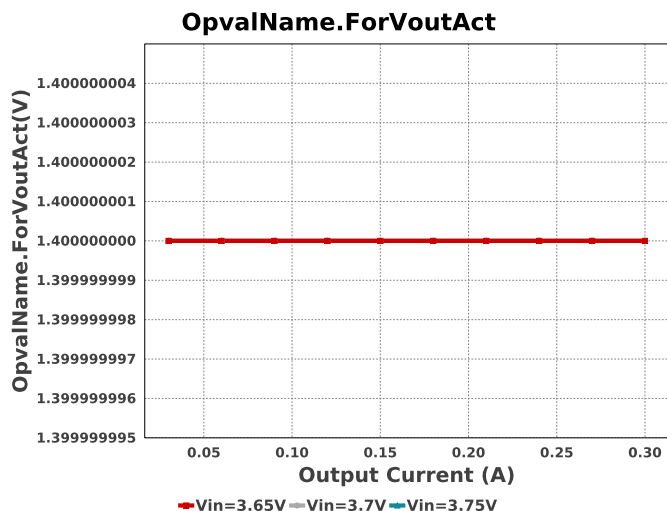
Design : 37 TPS7A4700RGWR
TPS7A4700RGWR 3.6V-3.75V to 3.30V @ 0.3A



1. This regulator device is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application. View WEBENCH(R) Disclaimer.

Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cin	MuRata	GRJ155R60J106ME11D Series= X5R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 0.0 A	1	\$0.02	 0402_070 3 mm ²
Cnr	MuRata	GRM188R60J105KA01D Series= X5R	Cap= 1.0 uF ESR= 6.065 mOhm VDC= 6.3 V IRMS= 1.36934 A	1	\$0.01	 0603 5 mm ²
Cout	MuRata	GRJ155R60J106ME11D Series= X5R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 0.0 A	1	\$0.02	 0402_070 3 mm ²
U1	Texas Instruments	TPS7A4700RGWR	Switcher	1	\$2.19	 RGW0020A 49 mm ²



Operating Values

#	Name	Value	Category	Description
1.	Output Noise RMS	4.273 μ V	General	Noise RMS
2.	IC Igroud	2.673 mA	IC	IC ground current
3.	IC Pd	145.023 mW	IC	IC power dissipation
4.	IC Tj	49.713 degC	IC	IC junction temperature
5.	IC Tolerance	35.0 mV	IC	IC Feedback Tolerance
6.	ICThetaJA	32.5 degC/W	IC	IC junction-to-ambient thermal resistance
7.	Iin Avg	302.67 mA	IC	Average input current
8.	IOUT_OP	300.0 mA	Op Point	Iout operating point
9.	Input Ripple Frequency	100.0 kHz	Op Point	Input Source Ripple Frequency for PSRR Calculation
10.	PSRR est.	-60.0 dB	Op Point	Power Supply Rejection Ratio estimated
11.	VIN_OP	3.75 V	Op Point	Vin operating point
12.	Total Pd	145.023 mW	Power	Total Power Dissipation
13.	Vout Act	3.3 V	Resistor	Achieved Vout with feedback resistor pair
14.	BOM Count	4	System Information	Total Design BOM count
15.	Efficiency	87.223 %	System Information	Steady state efficiency
16.	FootPrint	60.0 mm ²	System Information	Total Foot Print Area of BOM components
17.	Pout	990.0 mW	System Information	Total output power
18.	Total BOM	\$2.24	System Information	Total BOM Cost
19.	Vin p-p	40.0 mV	System Information	Input Source ripple voltage
20.	Vout	3.3 V	System Information	Operational Output Voltage
21.	Vout Tolerance	1.061 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
22.	Vout p-p	40.0 μ V	System Information	Peak-to-peak output ripple voltage

Appendix B. Bill of Quantities

Index	MPN	Manufacturer	Package	Description	Multiple	Unit Price(\$)	Extended Price(\$)
1	ADMSAIG-8.000MHZ-1Z-T	ABRACON	SMD3225-4P	±50ppm 18pF ±10ppm SMD3225-4P Crystals RoHS	1	1.0365	1.04
2	CRCW06032K20KFKEA	VISHAY		603 100mW 2.2kΩ 75V Thick Film Resistor ±1% ±100ppm/°C 0603 Chip Resistor - Surface Mount RoHS	100	0.0042	0.42
3	CGA3E2C0G1H102JT0Y0N	TDK		603 1nF ±5% 50V Ceramic Capacitor C0G 0603	10	0.0209	0.21
4	CRCW06031M00KFKEA	VISHAY		603 100mW 1MΩ 75V Thick Film Resistor ±1% ±100ppm/°C 0603 Chip Resistor - Surface Mount RoHS	100	0.0047	0.47
5	GCM1885C1H682JA16D	nuRata		603 6.8nF ±5% 50V Ceramic Capacitor C0G 0603	10	0.0503	0.5
6	ADS1292HRSMR	TI	VQFN-32(44)	SPI 2.7V 5.25V VQFN-32(44) Analog Front End (AFE) RoHS	1	5.7783	5.78
7	BLM21AG601SN1D	nuRata		805 600Ω±100MHz 1 Line Ferrite Bead 0805 700mA 210mΩ	20	0.0652	1.3
8	FRC0803F1002TS	FOJAN		805 10kΩ ±1% 125mW 0805 Thick Film Resistor	100	0.0022	0.22
9	GRM1885C1H3R0CA01D	nuRata		603 8pF 50V Ceramic Capacitor C0G 0603	50	0.0099	0.5
10	GCM1885C2A160JA16D	nuRata		603 16pF C0G ±5% 100V 0603 Ceramic Capacitors RoHS	50	0.0176	0.88
11	ABS07-32.768KHZ-6-T	ABRACON	SMD3215-2P	Crystal 32.768kHz ±20ppm 6pF 70kΩ SMD3215-2P	1	0.6692	1.34
12	GRM1555C1H72JE01D	nuRata		402 4.7nF ±5% 50V Ceramic Capacitor C0G 0402	50	0.0254	1.27
13	ABMM-8.000MHZ-B2-T	ABRACON	SMD7050-4P	Crystal 8MHz ±30ppm 18pF 70Ω SMD7050-4P	1	1.1772	1.18
14	RC0603JR-0710KTL	YAGEO		603 100mW 10kΩ 75V Thick Film Resistor ±100ppm/°C ±5% 0603 Chip Resistor - Surface Mount RoHS	100	0.0015	0.15
15	STM32F103C8T6	ST	LQFP-48(7x7)	ARM Cortex-M3 32 Bit 72MHz 37 LQFP-48(7x7) Microcontrollers RoHS	1	1.456	1.46
16	RC0603FR-0720RL	YAGEO		603 20Ω 100mW 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0017	0.17
17	RC06033F75R0TS	FOJAN		603 75Ω ±1% 100mW 0603 Thick Film Resistor	100	0.0012	0.12
18	FRC0603F1500TS	FOJAN		603 150Ω ±1% 100mW 0603 Thick Film Resistor	100	0.0013	0.13
19	ERA3AE8184V	PANASONIC		603 180kΩ ±0.1% 100mW 0603 Thin Film Resistor AEC-Q200	5	0.0932	0.47
20	AD832ACPZ-R7	ADI	LFCSP-20(44)	LFCSP-20(44) Specialized ICs RoHS	1	7.898	7.9
21	RC0603FR-07180KL	YAGEO		603 100mW 180kΩ 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0016	0.32
22	GRM1555C1H3R9CA01D	nuRata		402 3.9pF 50V Ceramic Capacitor C0G 0402	100	0.0044	0.44
23	GJM1555C1H6R0BD01D	nuRata		402 6pF 50V Ceramic Capacitor C0G 0402	50	0.0172	0.86
24	B5B-XB-AM(LF)(SN)	JST	Through Hole,P=2.5mm	Connector Header 5 position 2.5mm Pitch 3A Through Hole -25°C~+85°C	5	0.1146	0.57
25	RC0603FR-0710KTL	YAGEO		603 100mW 100kΩ 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0015	0.15
26	CL10A475KPNNNC	Samsung Electr		603 4.7nF ±10% 10V Ceramic Capacitor X5R 0603	50	0.0088	0.44
27	CL10A471KB8NNNC	Samsung Electr		603 470nF ±10% 50V Ceramic Capacitor X5R 0603	10	0.0184	0.18
28	RC0603JR-07150RL	YAGEO		603 100mW 150Ω 75V Thick Film Resistor ±100ppm/°C ±5% 0603 Chip Resistor - Surface Mount RoHS	100	0.0015	0.15
29	CL10B101KCN8NNC	Samsung Electr		603 100mF ±10% 100V Ceramic Capacitor X7R 0603	50	0.0077	0.39
30	CL05A105K05NNNC	Samsung Electr		402 1nF ±10% 16V Ceramic Capacitor X5R 0402	100	0.0038	0.38
31	CL05B101K05NNNC	Samsung Electr		402 100nF ±10% 16V Ceramic Capacitor X7R 0402	100	0.0017	0.17
32	GJM1555C1H2R7BD01D	nuRata		402 2.7pF 50V Ceramic Capacitor C0G 0402	50	0.0169	0.85
33	RC0603FR-0717ML	YAGEO		603 100mW 1MΩ 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0017	0.17
34	GRM1555C1H1R5CA01D	nuRata		402 1.5pF 50V Ceramic Capacitor C0G 0402	100	0.0035	0.35
35	RC0402FR-070RL	YAGEO		402 0Ω 62.5mW 50V Thick Film Resistor ±1% 0402 Chip Resistor - Surface Mount RoHS	100	0.0008	0.08
36	LQG15HS1NS02D	nuRata		402 1A 1.3nH 70mΩ Multilayer inductor 0402 Fixed Inductors RoHS	50	0.0231	1.16
37	LQG15HS1NS02D	nuRata		402 950mA 1.8nH 100mΩ Multilayer inductor 0402 Fixed Inductors RoHS	100	0.0117	1.17
38	LSMGA16080ST470RG	Taiyo Yuden		603 20mΩ 1 ±25% 470Ω/100MHz 0603 Ferrite Beads and Chips RoHS	5	0.0841	0.42
39	NX2016SA-32MHZ-ESX00A-C-NDK		SMD2016-4P	Crystal 32MHz 8pF 60Ω SMD2016-4P	1	0.4352	0.44
40	NX3215SA-32.768K-STD-MU-ANDK		SMD3215-2P	Crystal 32.768kHz ±20ppm 6pF 70kΩ SMD3215-2P	5	0.2831	1.42
41	2450AT18A100E	Johanson Dielec		1206 RF ANTENNA 2.4G Surface Mount 1206	1	0.5287	0.53
42	STM32WBAS2K6UG6	ST	UFQFPN-32(5x5)	UFQFPN-32(5x5) Microcontrollers RoHS	1	4.5301	4.53
43	RT06033BRD074K12L	YAGEO		603 4.12kΩ 100mW 75V Thin Film Resistor ±25ppm/°C ±0.1% 0603 Chip Resistor - Surface Mount RoHS	20	0.0261	0.52
44	CL21A106KYQNNNE	Samsung Electr		805 10nF ±10% 50V Ceramic Capacitor X5R 0805	5	0.0727	0.73
45	CL21B105KPFNNNE	Samsung Electr		805 1uF ±10% 10V Ceramic Capacitor X7R 0805	50	0.0117	0.59
46	CL21A226KPLRNC	Samsung Electr		805 22nF ±10% 10V Ceramic Capacitor X5R 0805	10	0.0907	0.91
47	LQM2HPN2R2MG0L	nuRata		1008 1.3A 2.2uH ±20% 100mΩ 1008 Fixed Inductors RoHS	5	0.1888	0.94
48	ERA-3AE8251V	PANASONIC		603 8.2kΩ ±0.1% 100mW 0603 Thin Film Resistor AEC-Q200	5	0.0752	0.38
49	ERJP16B1871V	PANASONIC		805 250mW 1.87kΩ Thick Film Resistor ±50ppm/°C ±0.1% 0805 Chip Resistor - Surface Mount RoHS	5	0.092	0.46
50	CL10B101KB8NNNC	Samsung Electr		603 100mF ±10% 50V Ceramic Capacitor X7R 0603	100	0.0066	0.66
51	CL21B475KOFNNNE	Samsung Electr		805 4.7nF ±10% 16V Ceramic Capacitor X7R 0805	10	0.0334	0.33
52	CL21B105KOFNNNE	Samsung Electr		805 1uF ±10% 16V Ceramic Capacitor X7R 0805	20	0.015	0.3
53	RT06033BRD071K18L	YAGEO		603 100mW 1.18kΩ 75V Thin Film Resistor ±25ppm/°C ±0.1% 0603 Chip Resistor - Surface Mount RoHS	20	0.0256	0.51
54	MAX1704SC -T10	MAXIM	DFN-8-EP(2x2)	Lithium Battery I2C DFN-8-EP(2x2) Battery Management RoHS	1	2.1833	4.37
55	RT06033BRD071K19L	YAGEO		603 100mW 1.18kΩ 75V Thin Film Resistor ±25ppm/°C ±0.1% 0603 Chip Resistor - Surface Mount RoHS	20	0.0262	0.52
56	RC0603FR-071K5L	YAGEO		603 100mW 1.5kΩ 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0017	0.17
57	NTCLE100E3103JBD0	VISHAY	Through Hole,P=2.54mm	NTC Thermistor 10kΩ Through Hole,P=2.54mm	1	0.5317	0.53
58	RC0603JR-071KLL	YAGEO		603 100mW 1kΩ 75V Thick Film Resistor ±100ppm/°C ±5% 0603 Chip Resistor - Surface Mount RoHS	100	0.0016	0.16
59	RC0805JR-071KLL	YAGEO		805 125mW 1kΩ 150V Thick Film Resistor ±100ppm/°C ±5% 0805 Chip Resistor - Surface Mount RoHS	100	0.0026	0.26
60	TYPE-C 16PIN 2MD(073)	SHOU HAN	SMD	USB-C (USB TYPE-C) Receptacle Connector 16 Position Surface Mount, Right Angle	20	0.0705	1.41
61	C0805C105K8RACTU	KEMET		805 1uF X7R ±10% 10V 0805 Ceramic Capacitors RoHS	20	0.0353	0.71
62	FRC0805F0000TS	FOJAN		805 125mW 0Ω 150V Thick Film Resistor ±400ppm/°C ±1% 0805 Chip Resistor - Surface Mount RoHS	100	0.0023	0.23
63	CRCW0603000020EA	VISHAY		603 0Ω 100mW 100V Thick Film Resistor 0603 Chip Resistor - Surface Mount RoHS	100	0.0028	0.28
64	YZA-009	YIZHI	-	Rectangle button SPST Tactile Switches	20	0.0171	0.34
65	RT06033BRD07100KL	YAGEO		603 100kΩ 100mW 75V Thin Film Resistor ±25ppm/°C ±0.1% 0603 Chip Resistor - Surface Mount RoHS	20	0.0236	0.47
66	RT08053BRD07332KL	YAGEO		805 125mW 332kΩ 150V Thin Film Resistor ±25ppm/°C ±0.1% 0805 Chip Resistor - Surface Mount RoHS	20	0.0421	0.84
67	RT08053BRD07360KL	YAGEO		805 125mW 360kΩ 150V Thin Film Resistor ±25ppm/°C ±0.1% 0805 Chip Resistor - Surface Mount RoHS	20	0.0389	0.78
68	GRM1885C1H272JA01D	nuRata		603 2.7nF ±5% 50V Ceramic Capacitor C0G 0603	50	0.0253	1.27
69	GRM1885C1H682JA01D	nuRata		603 6.8nF ±5% 50V Ceramic Capacitor C0G 0603	50	0.0242	1.21
70	CC0603FRNPO8B102	YAGEO		603 1nF ±1% 50V Ceramic Capacitor NP0 0603	50	0.0147	0.74
71	CC0603KRX7R8B334	YAGEO		603 330nF ±10% 25V Ceramic Capacitor X7R 0603	50	0.0128	0.64
72	GRM1885C1H103JA01D	nuRata		603 10nF ±5% 50V Ceramic Capacitor C0G 0603	20	0.0446	0.89
73	FRC0603F14QTS	FOJAN		603 1.4MΩ ±1% 100mW 0603 Thick Film Resistor	100	0.0015	0.15
74	TPS63900DSKR	TI	WS0N-10-EP(2.5x2.5)	Back-Boost Adjustable 1.8V~5V 400mA WS0N-10-EP(2.5x2.5) Voltage Regulators - DC DC Switching Re	1	1.26	3.78
75	LIS3DHTR	ST	LGA-16(3x3)	LGA-16(3x3) Accelerometers RoHS	1	1.015	2.03
76	PJ-320B	SOFNG	SMD	3.5mm Headphone Jack 25mA 24V Surface Mount	1	0.0393	0.39
77	RC0603FR-0710ML	YAGEO		603 100mW 10MΩ 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0015	0.15
78	RC0603FR-075KLL	YAGEO		603 100mW 5.1kΩ 75V Thick Film Resistor ±100ppm/°C ±1% 0603 Chip Resistor - Surface Mount RoHS	100	0.0018	0.18
79	RC0805FR-071K5L	YAGEO		805 125mW 1.5kΩ 150V Thick Film Resistor ±100ppm/°C ±1% 0805 Chip Resistor - Surface Mount RoHS	100	0.0029	0.29
80	UJ20-C-H-G-SMT-1-P16-TR	CUI	SMD	USB 2.0 SMD USB, DVI, HDMI Connector Adapters RoHS	1	0.4255	0.85
81	MK-12D12G1-B	HOOYA	Through Hole,P=2.5mm	Slide Switch 300mA @ 12V Through Hole	5	0.0892	0.45
82	BQ24074RC7R	TI	QFN-16-EP(3x3)	4.35V~10.2V Lithium Battery 1 1.5A QFN-16-EP(3x3) Battery Chargers ICs RoHS	1	2.5507	5.1
83	TPS7A4700RGWR	TI	VQFN-20-EP(5x5)	1.4V~20.5V Positive Adjustable VQFN-20-EP(5x5) Voltage Regulators - Linear, Low Drop Out (LDO) Re	1	3.474	6.95

Appendix C. ECG Comparative Data

This appendix presents the side-by-side signal evaluation between the custom-designed single-lead ECG prototype and a gold-standard clinical asset.

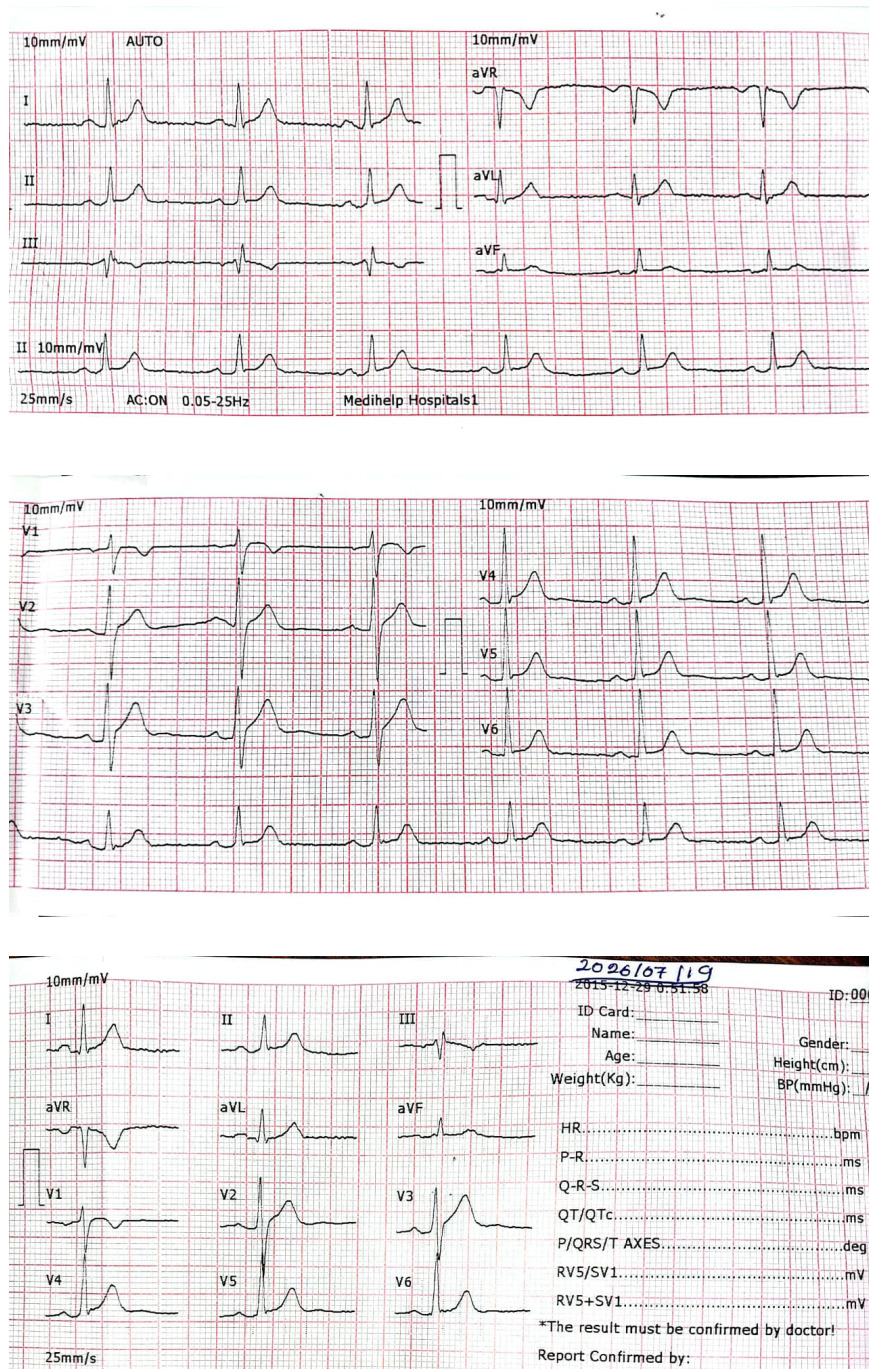


Figure 15: Full sequential 12-lead hospital ECG printout for subject R. M. S. H. Ratnayake (recorded 14/07/2026, 7:45 p.m. at Medihelp Hospitals, Moratuwa; Ref. No. 777), utilized as the baseline clinical benchmark for device comparison.

Device: BT05

Recording ID: 1784533631775

Metric	Value
Duration	0:23
Average Heart Rate	83 bpm
Paper Speed / Scale	25 mm/s (1 mm = 0.04s)
Signal Quality	Good

Recorded ECG Waveform (Standard 25 mm/s Scale)

Strip 1 (0.0s - 7.5s)



Strip 2 (7.5s - 15.0s)



Strip 3 (15.0s - 22.4s)



Strip 4 (22.4s - 23.0s)

